COEP Technological University

(COEP Tech)
A Unitary Public University of Government of Maharashtra
w.e.f 21st June 2022
(Formerly College of Engineering Pune)

YEAR : SECOND YEAR
SEMISTER: III

SUBJECT: Geo-informatics

Prepared by,
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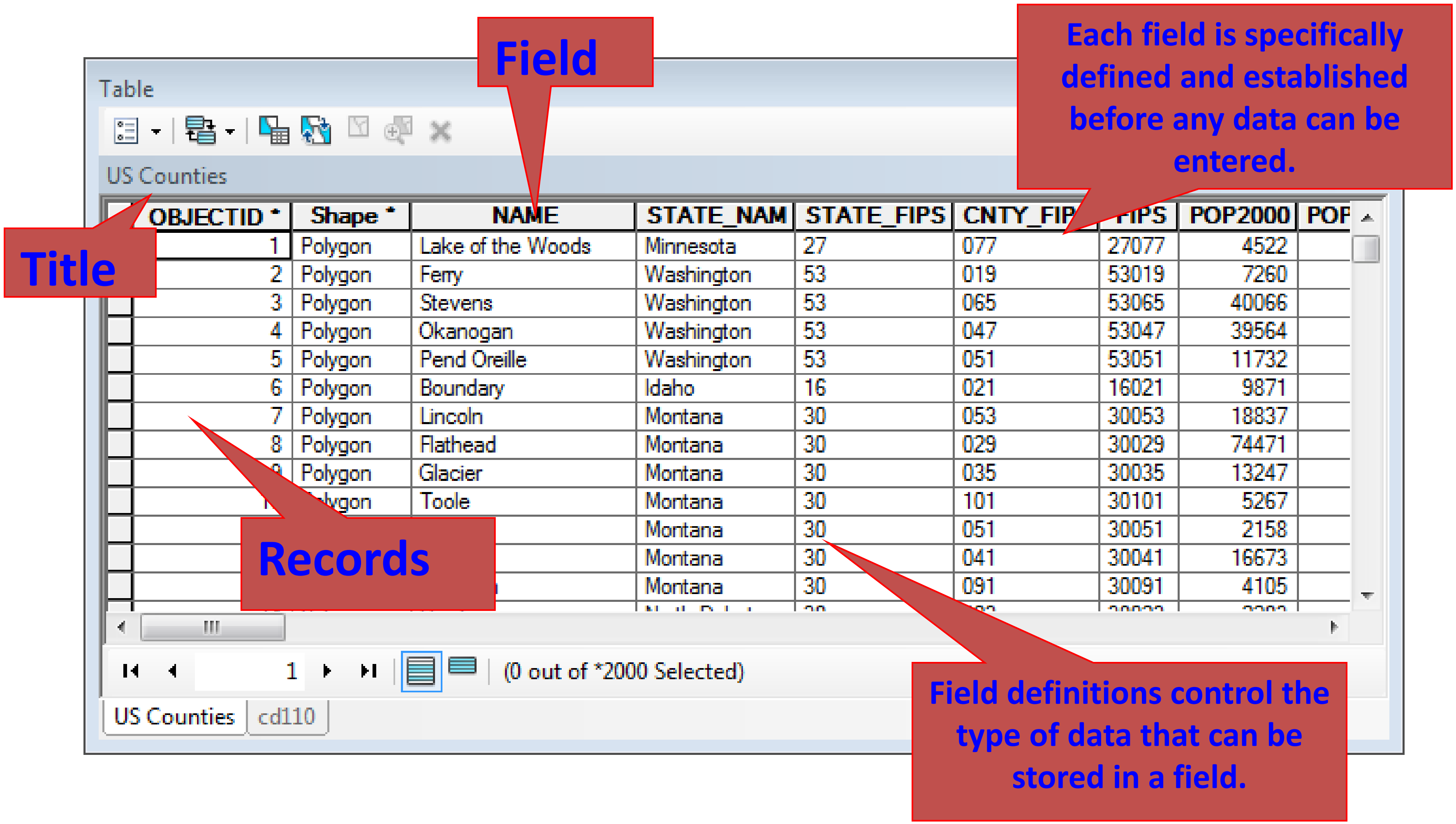
Attribute Data & Tables

Table terminology



Types of tables

- Attribute table
  - Stores attributes of map features
  - Associated with a spatial data layer
  - Has special fields for spatial information
- Standalone table
  - Stores any tabular data
  - Not associated with spatial data
  - OID instead of FID

US Counties

OBJ	Shape *	NAME	STATE_NAM
1	Polygon	Lake of the Woods	Minnesota
2	Polygon	Ferry	Washington
3	Polygon	Stevens	Washington
4	Polygon	Okanogan	Washington
5	Polygon	Pend Oreille	Washington
6	Polygon	Boundary	Idaho

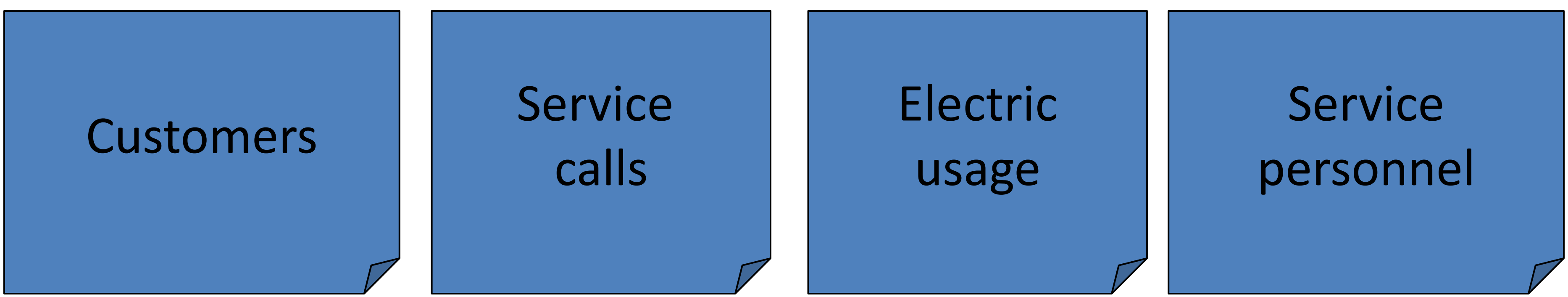
NE_reps

OID	OBJECTID	NAME	PARTY	STATE_
0	169	Chellie Pingree	Democrat	ME
1	341	James A. Himes	Democrat	CT
2	343	Christopher S. Murphy	Democrat	CT
3	370	James P. McGovern	Democrat	MA
4	371	Joe Courtney	Democrat	CT
5	372	Patrick J. Kennedy	Democrat	RI

Database Management Systems

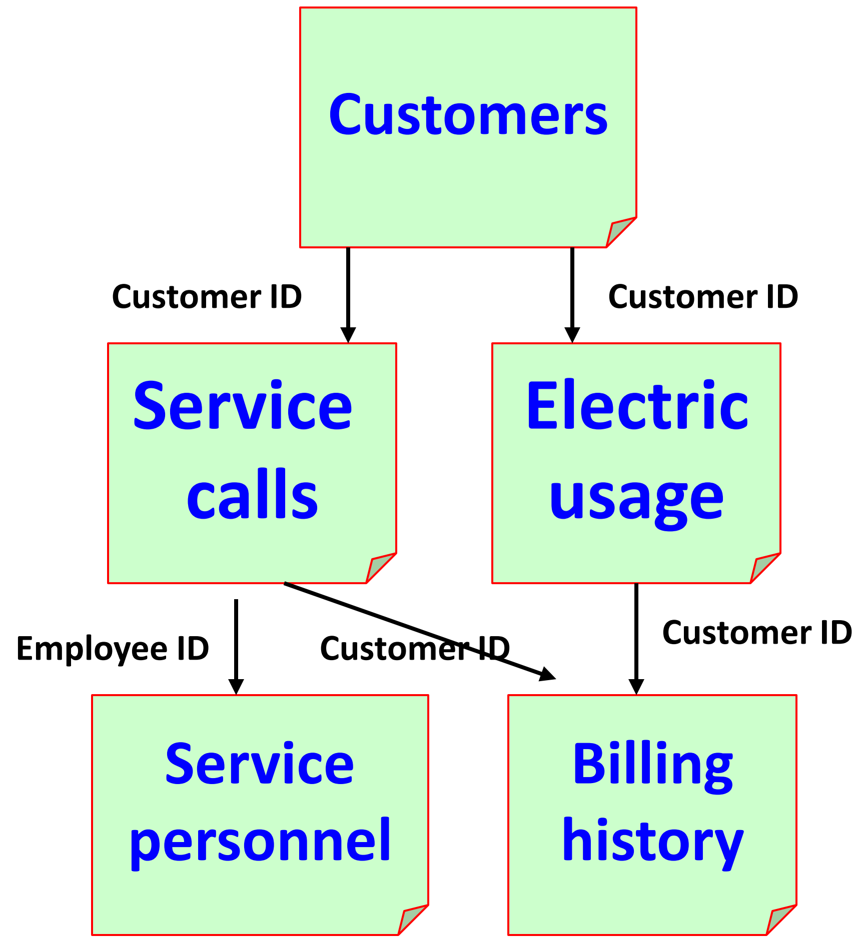
Flat file DBMS

- Flat file
  - Stores data as rows of information in files
  - Simple and robust
  - Inefficient for search and query

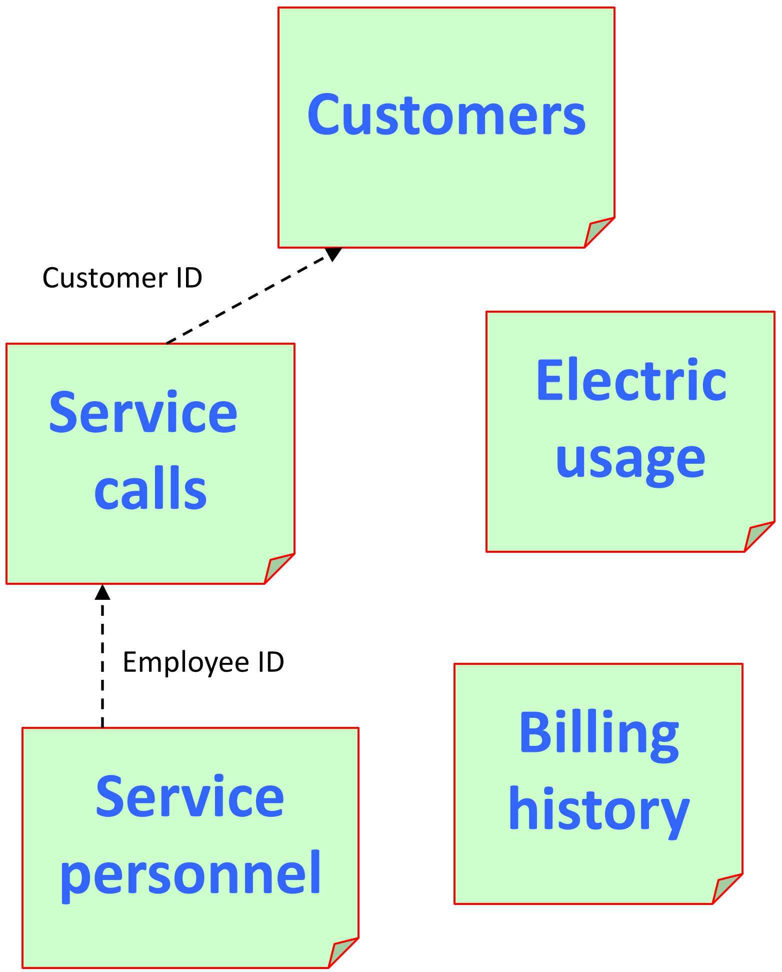


Hierarchical DBMS

- Stores data in multiple tables
- Tables have defined parent-child relationships
- Pre-set hierarchy of table relationships designed for specific queries
- Very efficient for specific queries
- Range of queries limited by structure



Relational DBMS



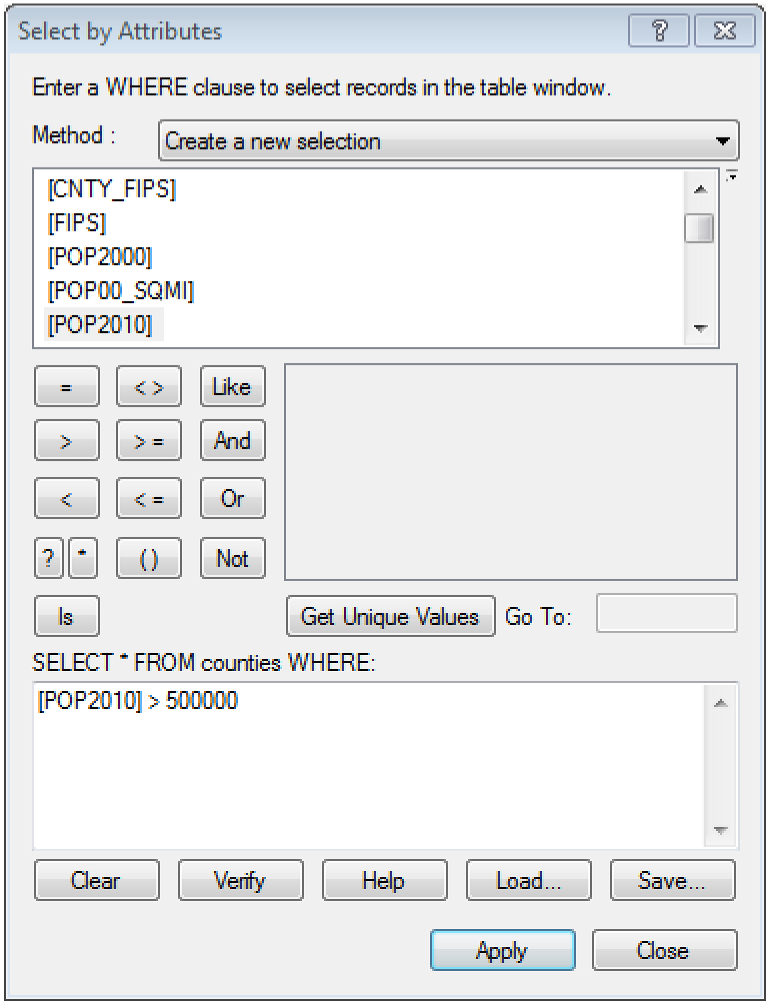
Queries- What is a query?

- A query extracts certain records from a table based on a specified condition
- For example,
  - How many students have GPA > 3.5?
  - How many states have population > 1 million?
  - How many counties had greater population in 1990 than in 2000?
  - How many customers have last names beginning with “Mac” or “Mc”?

Structured Query Language, or SQL

- Many databases use a special query language called Structured Query Language
- Can write queries that work in multiple DBMS environments
- Queries can be saved and reused
- Nearly always case-sensitive

SQL Query Examples



Some Valid Queries

SELECT *FROM cities WHERE
"POP1990" >= 500000

SELECT *FROM counties WHERE
“BEEFCOW_92” < “BEEFCOW_87”

SELECT *FROM parcels WHERE
“LU-CODE” = 42 AND “VALUE” > 50000

SELECT *FROM rentals WHERE
“RENT” > 700 AND “RENT” < 1500

In most databases, SQL expressions are case-sensitive “Smith” ≠ “SMITH”

Multiple Criteria Queries

- Multiple criteria queries using AND or OR

  “STATE” = ‘Alabama’ OR “STATE” = ‘Texas’

  [Value] > 5000 AND [Value] < 10000

  - Note that the field name must be repeated for each condition

Searching for partial matches

- Sometimes you need to find one string within another rather than an exact match
  - Find all customer names beginning with "Mac" or "Mc"
  - Find all zip codes beginning with 0
- Typically uses a "wildcard" character "*"
  - *Mac* or *Mc*
  - 0*

The Like Operator

- “NAME” LIKE ‘%(D)%’
  - Finds all of the (D) Democrats
- % is the “IGNORE” wildcard
- Ignores Don or Danforth

NAME
Sonny Callahan (R)
Terry Everett (R)
Bob Riley (R)
Robert B. Aderholt (R)
Robert E. "Bud" Cramer, Jr. (D)
Spencer Bachus (R)
Earl F. Hilliard (D)
Don Young (R)
Matt Salmon (R)
Ed Pastor (D)
Bob Stump (R)
John Shadegg (R)
Jim Kolbe (R)

- “NAME” LIKE ‘%New %’
  - Would find New Hampshire and New York, but not Newcastle or Kennewick

NAME
Mesa
Sun City West
Cottonwood-Verde Village
New River
Prescott
Prescott Valley
Flagstaff
Sedona
Payson
Cedar City
Green River
Rock Springs
Magna

Query results

- The result of a query is a set of selected records that meet the criteria
- Subsequent operations on the table will use only the selected records and ignore the others. A user might…
  - Export the selected records to a new table
  - Calculate statistics on the selected records
  - Calculate new values for the selected records
  - Produce a report based on the selected records

Joining tables

US Counties

OBJ	Shape *	NAME	STATE_NAM	FIPS
1	Polygon	Lake of the Woods	Minnesota	27077
2	Polygon	Ferry	Washington	53019
3	Polygon	Stevens	Washington	53065
4	Polygon	Okanogan	Washington	53047
5	Polygon	Pend Oreille	Washington	53051
6	Polygon	Boundary	Idaho	16021
7	Polygon	Lincoln	Montana	30[illegible]53

popestmt00

OBJECTID *	FIPS	POP2002	POP2001	POP2000
1	01001	45604	44698	43903
2	01003	147932	144787	141410
3	01005	28826	28993	29047
4	01007	21838	21935	20869
5	01009	52968	52143	51213
6	01011	11367	11454	11613
7	[illegible]1013	20911	21157	21336

US Counties

OBJ	Shape *	NAME	STATE_NAM	FIPS	OBJE	FIPS	POP2002	POP2001	POP2000
1	Polygon	Lake of the Woods	Minnesota	27077	1351	27077	4385	4460	4507
2	Polygon	Ferry	Washington	53019	2962	53019	7268	7290	7292
3	Polygon	Stevens	Washington	53065	2985	53065	40556	40477	40246
4	Polygon	Okanogan	Washington	53047	2976	53047	39186	39303	39571
5	Polygon	Pend Oreille	Washington	53051	2978	53051	12008	11861	11745
6	Polygon	Boundary	Idaho	16021	560	16021	10085	9946	9924
7	Polygon	Lincoln	Montana	30053	1623	30053	[illegible]	18691	18840
8	Polygon	Flathead	Montana	30029	1611	30029	[illegible]	[illegible]	[illegible]

Joined table

Join facts

- Joins are temporary relationships between tables used by a relational DBMS
- Tables must share a common field (key)
- Treats the two tables as a single table
- Original stored data is not affected
- Can be removed when no longer needed

One-to-one joins

Attributes of US States

FID	Shape*	STATE_NAME	STATE_ABBR
0	Polygon	Hawaii	HI
1	Polygon	Washington	WA
2	Polygon	Montana	MT
3	Polygon	Maine	ME
4	Polygon	North Dakota	ND
5	Polygon	South Dakota	SD
6	Polygon	Wyoming	WY
7	Polygon	Wisconsin	WI

quakesum

STATE	Count_	Sum_DAMAGE	Su
CA	218	3705234000	
AK	106	32600000	
MT	62	4220000	
WA	67	3775000	
ID	41	1350000	
HI	63	1100000	
OR	24	760000	

Types of Cardinality

Rule of Joining

Attributes of US States

FID	Shape*	STATE_NAME	STATE_ABBR
0	Polygon	Hawaii	HI
1	Polygon	Washington	WA
2	Polygon	Montana	MT
3	Polygon	Maine	ME
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5	Polygon	South Dakota	SD
6	Polygon	Wyoming	WY
7	Polygon	Wisconsin	WI

quakesum

STATE	Count_	Sum_DAMAGE	Sum_DEA
CA	218	3705234000	
AK	106	32600000	
MT	62	4220000	
WA	67	3775000	
ID	41	1350000	
HI	63	1100000	
OR	24	760000	

Destination table

Source table

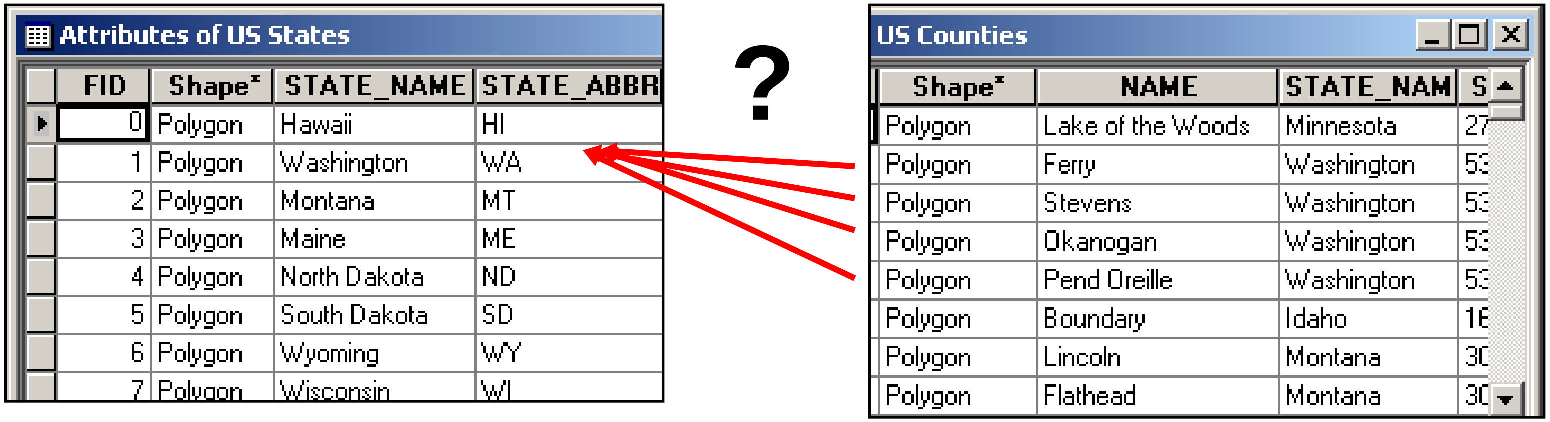
US Counties

Shape*	NAME	STATE_NAM	S
Polygon	Lake of the Woods	Minnesota	27
Polygon	Ferry	Washington	53
Polygon	Stevens	Washington	53
Polygon	Okanogan	Washington	53
Polygon	Pend Oreille	Washington	53
Polygon	Boundary	Idaho	16
Polygon	Lincoln	Montana	30
Polygon	Flathead	Montana	30

Attributes of US States

FID	Shape*	STATE_NAME	STATE_ABBR
0	Polygon	Hawaii	HI
1	Polygon	Washington	WA
2	Polygon	Montana	MT
3	Polygon	Maine	ME
4	Polygon	North Dakota	ND
5	Polygon	South Dakota	SD
6	Polygon	Wyoming	WY
7	Polygon	Wisconsin	WI

One to many



Attributes of US States

FID	Shape*	STATE_NAME	STATE_ABBR
0	Polygon	Hawaii	HI
1	Polygon	Washington	WA
2	Polygon	Montana	MT
3	Polygon	Maine	ME
4	Polygon	North Dakota	ND
5	Polygon	South Dakota	SD
6	Polygon	Wyoming	WY
7	Polygon	Wisconsin	WI

?

US Counties

Shape*	NAME	STATE_NAM	S
Polygon	Lake of the Woods	Minnesota	27
Polygon	Ferry	Washington	53
Polygon	Stevens	Washington	53
Polygon	Okanogan	Washington	53
Polygon	Pend Oreille	Washington	53
Polygon	Boundary	Idaho	16
Polygon	Lincoln	Montana	30
Polygon	Flathead	Montana	30

Destination table

Source table

Violates the Rule of Joining

Record to join to destination is ambiguous

Must use a relate instead

No matches?

- Join still occurs, but <Null> values are given to records with no match

Earthquakes table has records for Puerto Rico

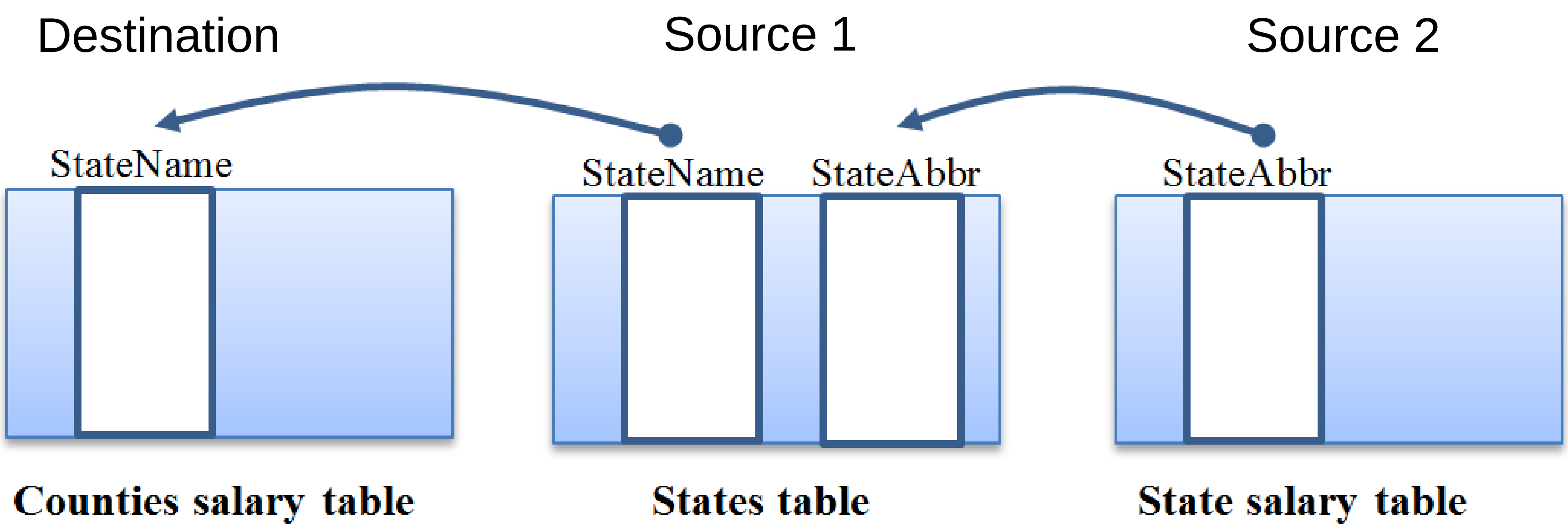
States table does not have records for Puerto Rico

quakehis

STA	LOCATION	STATE_	STATE	POP201
PR	Puerto Rico Region	<Null>	<Null>	<Null>
PR	Puerto Rico Region	<Null>	<Null>	<Null>
PR	Northwest of Puerto Ri	<Null>	<Null>	<Null>
AL	Irondale, Alabama	Alabama	AL	473559
AL	Near Birmingham, Alab	Alabama	AL	473559
AL	Near Birmingham, Alab	Alabama	AL	473559

Multiple joins

- You can add multiple joins to a destination table using different keys
- You cannot join to a joined source table

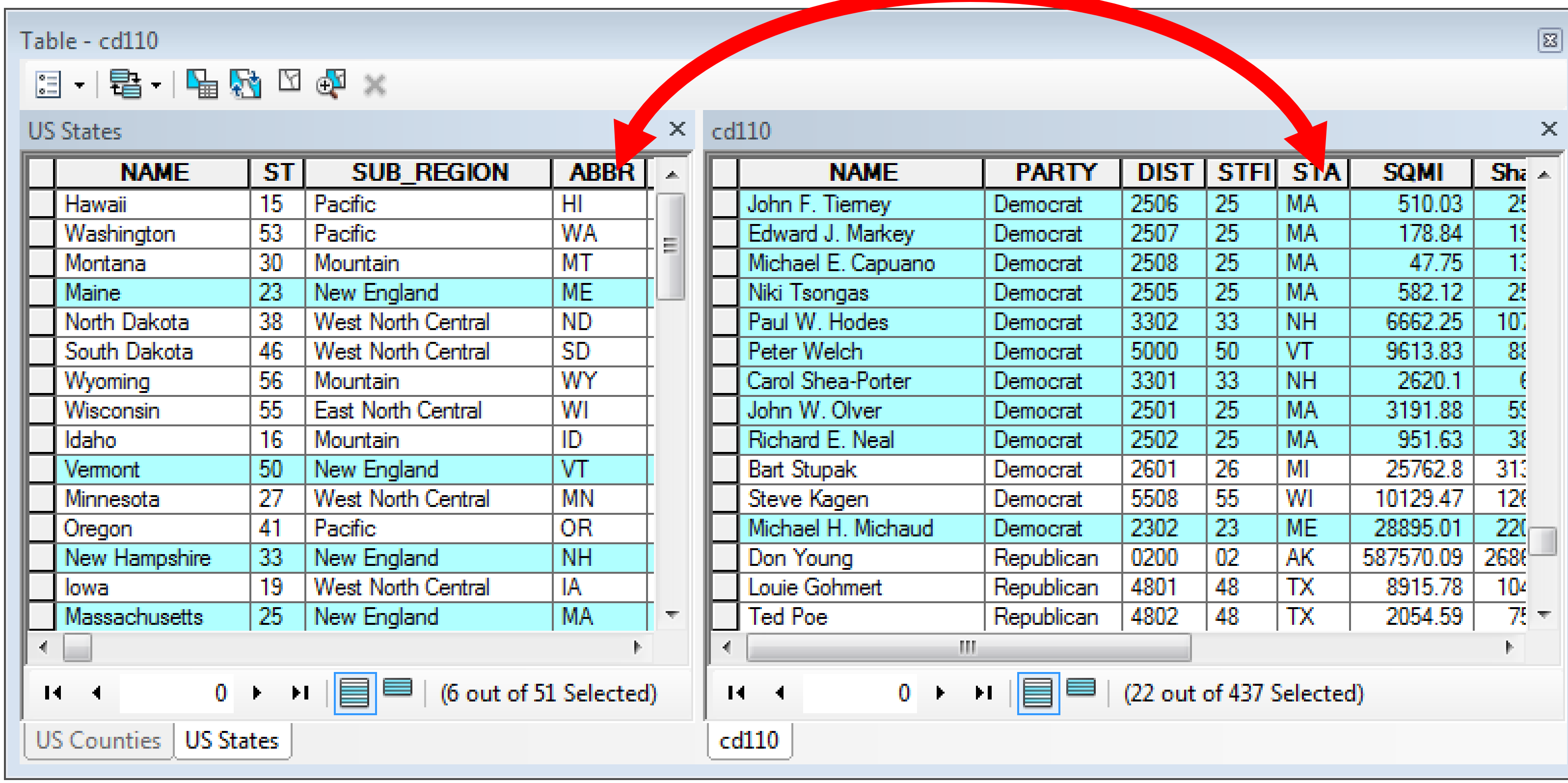


Relates

- Similar to a join except that
  - The tables remain separate
  - Items selected in one table may be highlighted in the related table

Related tables

Select the New England states



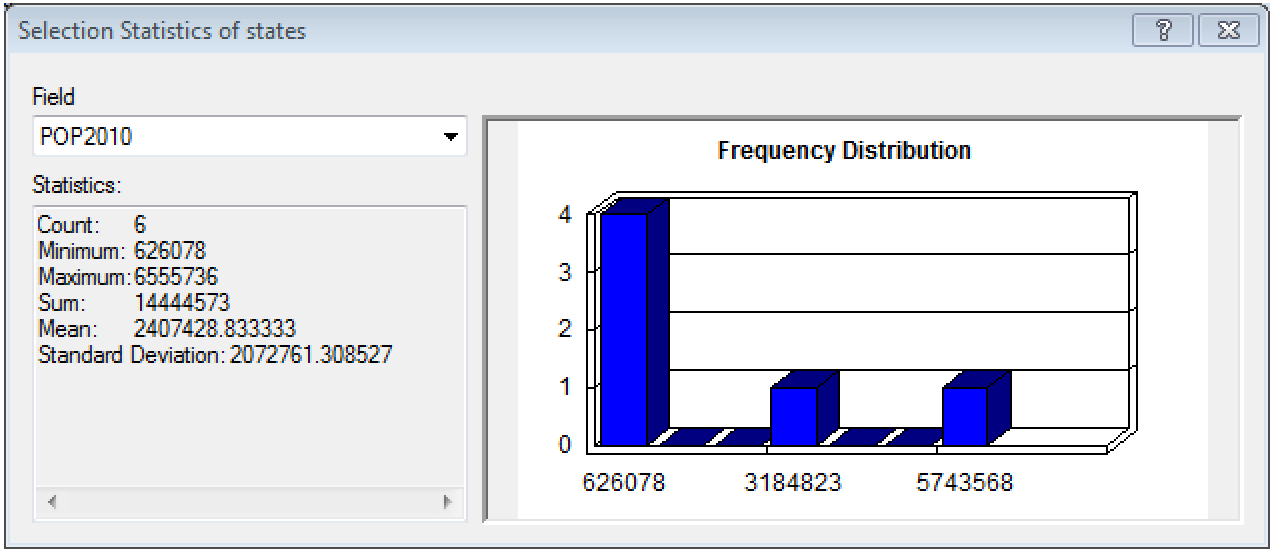
NAME	ST	SUB_REGION	ABBR
Hawaii	15	Pacific	HI
Washington	53	Pacific	WA
Montana	30	Mountain	MT
Maine	23	New England	ME
North Dakota	38	West North Central	ND
South Dakota	46	West North Central	SD
Wyoming	56	Mountain	WY
Wisconsin	55	East North Central	WI
Idaho	16	Mountain	ID
Vermont	50	New England	VT
Minnesota	27	West North Central	MN
Oregon	41	Pacific	OR
New Hampshire	33	New England	NH
Iowa	19	West North Central	IA
Massachusetts	25	New England	MA

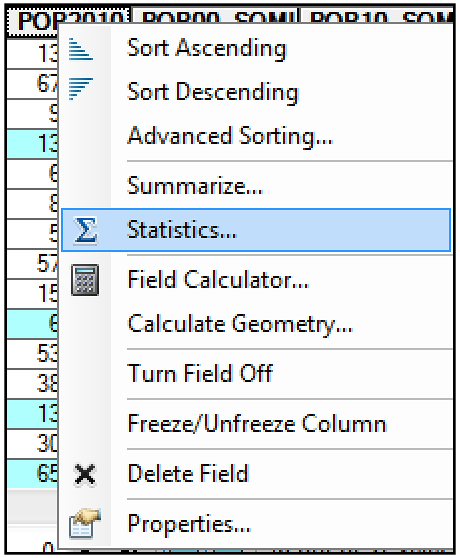
NAME	PARTY	DIST	STFI	STA	SQMI	Sha
John F. Tierney	Democrat	2506	25	MA	510.03	25
Edward J. Markey	Democrat	2507	25	MA	178.84	19
Michael E. Capuano	Democrat	2508	25	MA	47.75	13
Niki Tsongas	Democrat	2505	25	MA	582.12	25
Paul W. Hodes	Democrat	3302	33	NH	6662.25	107
Peter Welch	Democrat	5000	50	VT	9613.83	88
Carol Shea-Porter	Democrat	3301	33	NH	2620.1	6
John W. Olver	Democrat	2501	25	MA	3191.88	59
Richard E. Neal	Democrat	2502	25	MA	951.63	38
Bart Stupak	Democrat	2601	26	MI	25762.8	313
Steve Kagen	Democrat	5508	55	WI	10129.47	126
Michael H. Michaud	Democrat	2302	23	ME	28895.01	220
Don Young	Republican	0200	02	AK	587570.09	2686
Louie Gohmert	Republican	4801	48	TX	8915.78	104
Ted Poe	Republican	4802	48	TX	2054.59	75

Find reps of New England states

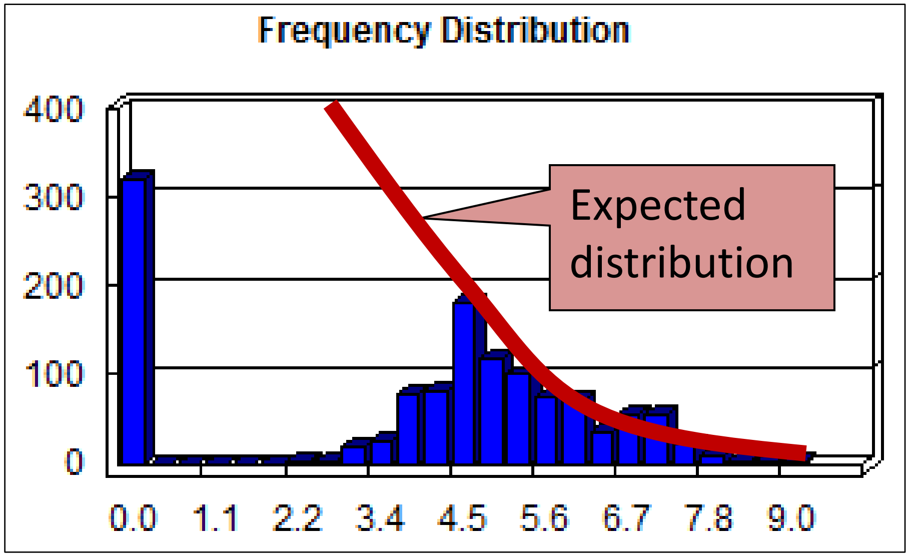
Table statistics

- Often a first step in data analysis
- Opportunity to think critically about data
- Returns statistics only for selected set





Thinking critically



Field

MAG

Statistics:

Count: 1264
Minimum: 0
Maximum: 9.23
Sum: 5006.140002
Mean: 3.960554
Standard Deviation: 2.507
Nulls: 0

- Why are there so many zero magnitude quakes?
- Why so few small earthquakes?
- What are the implications for data analysis?

Summarizing tables

- Calculate statistics for **groups** of features in a table
- Groups by unique values in the one field
- User chooses statistics to calculate for other fields
- Produces another table as output with groups and stats

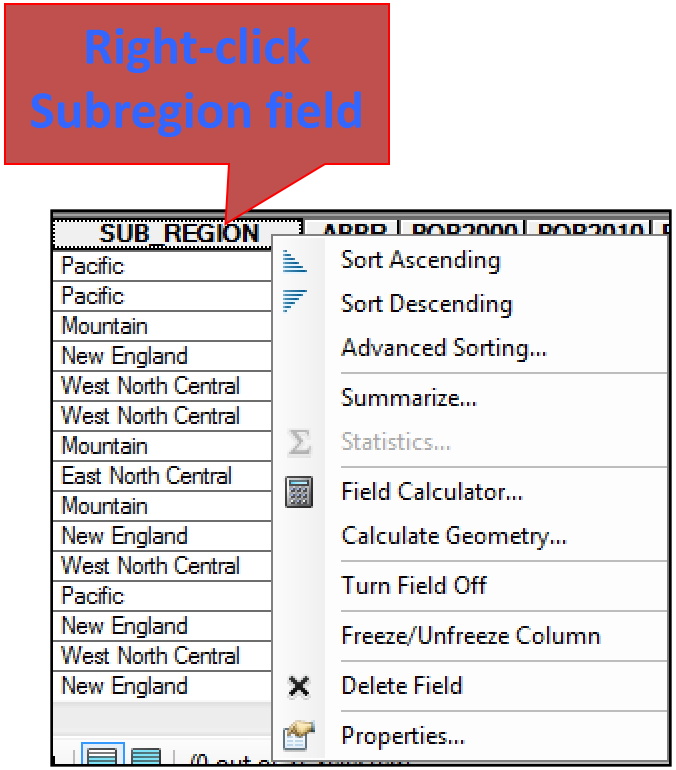
US States

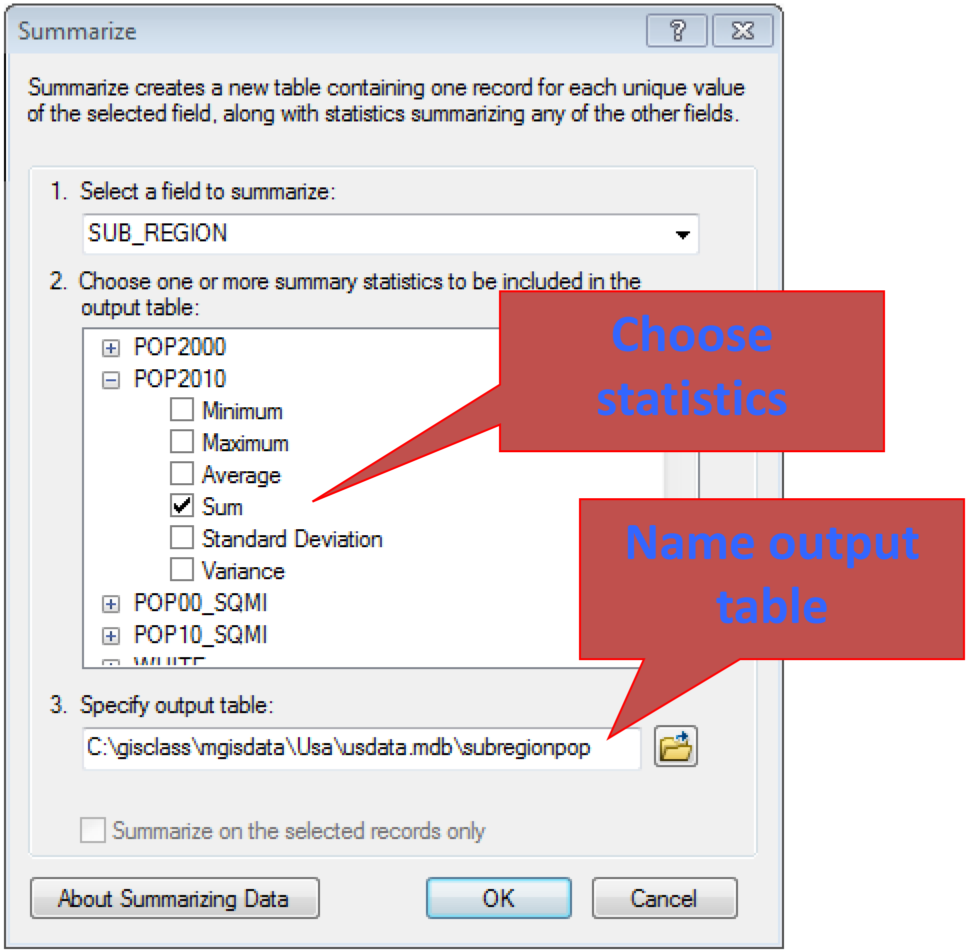
NAME	ST	SUB_REGION	ABBR	POP2000	POP2010
Hawaii	15	Pacific	HI	1211537	1309580
Washington	53	Pacific	WA	5894121	6756150
Montana	30	Mountain	MT	902195	983932
Maine	23	New England	ME	1274923	1338645
North Dakota	38	West North Central	ND	642200	662194
South Dakota	46	West North Central	SD	754844	827263
Wyoming	56	Mountain	WY	493782	548154
Wisconsin	55	East North Central	WI	5363675	5741617
Idaho	16	Mountain	ID	1293953	1581697
Vermont	50	New England	VT	608827	626078
Minnesota	27	West North Central	MN	4919479	5334772
Oregon	41	Pacific	OR	3421399	3865839
New Hampshire	33	New England	NH	1235786	1329915
Iowa	19	West North Central	IA	2926324	3057995

How many people live in each subregion?

What is the total area of each subregion?

How to summarize





Sum Pop2010

Sum SQMI

Summarize Output Table

subregionpop

	OBJECTID *	SUB_REGION	Count_SUB_REGION	Sum_POP2010
▸	1	East North Central	5	47020813
	2	East South Central	4	18438179
	3	Middle Atlantic	3	40940511
	4	Mountain	8	22621196
	5	New England	6	14444573
	6	Pacific	5	50611268
	7	South Atlantic	9	60166524
	8	West North Central	7	20549764
	9	West South Central	4	36420035

Count field always generated automatically

Create map

- Could we now create a map of population in subregions? No, there are no features.

Standalone table

subregionpop

OBJECTID *	SUB_REGION	Count_SUB_REGION	Sum_POP2010
1	East North Central	5	47020813
2	East South Central	4	18438179
3	Middle Atlantic	3	40940511
4	Mountain	8	22621196
5	New England	6	14444573
6	Pacific	5	50611268
7	South Atlantic	9	60166524
8	West North Central	7	20549764
9	West South Central	4	36420035

Joining the table

States layer

US States

OBJ	Shape *	NAME	ST	SUB_REGION
1	Polygon	Hawaii	15	Pacific
2	Polygon	Washington	53	Pacific
3	Polygon	Montana	30	Mountain
4	Polygon	Maine	23	New England
5	Polygon	North Dakota	38	West North Central
6	Polygon	South Dakota	46	West North Central
7	Polygon	Wyoming	56	Mountain
8	Polygon	Wisconsin	55	East North Central
9	Polygon	Idaho	16	Mountain
10	Polygon	Vermont	50	New England
11	Polygon	Minnesota	27	West North Central
12	Polygon	Oregon	41	Pacific
13	Polygon	New Hampshire	33	New England

Summarized table

subregionpop

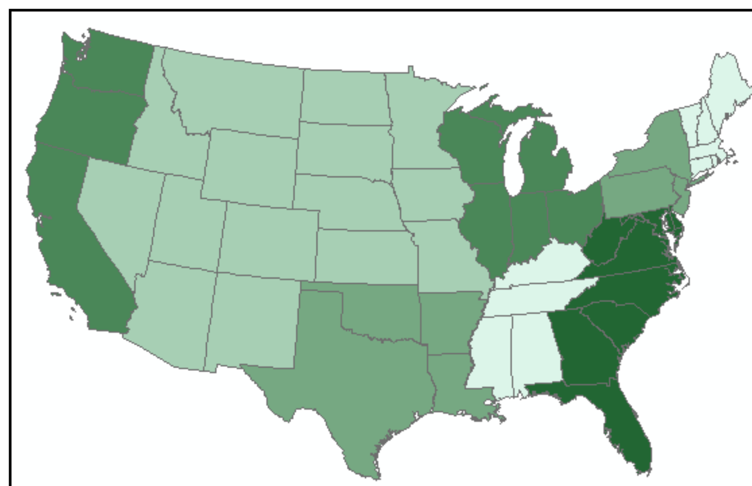
OBJECTID *	SUB_REGION	Count_SUB_REGION	Sum_POP2010
1	East North Central	5	47020813
2	East South Central	4	18438179
3	Middle Atlantic	3	40940511
4	Mountain	8	22621196
5	New England	6	14444573
6	Pacific	5	50611268
7	South Atlantic	9	60166524
8	West North Central	7	20549764
9	West South Central	4	36420035

Fields

Value: Sum_POP2010

Normalization: none

Color Ramp:



Database storage concepts

Fields

- Fields have specific types available
- Must be defined before use
- Once defined, cannot be changed
- Naming rules
  - No more than 13 characters
  - Use only letters and numbers
  - Must start with a letter

ArcGIS field data types

Field Type	Explanation	Examples
Short	Integers stored as 2-byte binary numbers *Range of values −32,000 to +32,000*	255 12001
Long	Integers stored as 10-byte binary numbers *Range of values −2.14 billion to +2.14 billion*	156000 457890
Float	Floating-point values with eight significant digits in the mantissa	1.289385e12 1.5647894e–02
Double	Double-precision floating-point values with 16 significant digits in the mantissa	1.12114118119141e13
Text	Alphanumeric strings	'Maple St' 'John H. Smith'
Date	Date format	07/12/92 10/17/63
BLOB*	Binary large object; any complex binary data including images, documents, etc.	

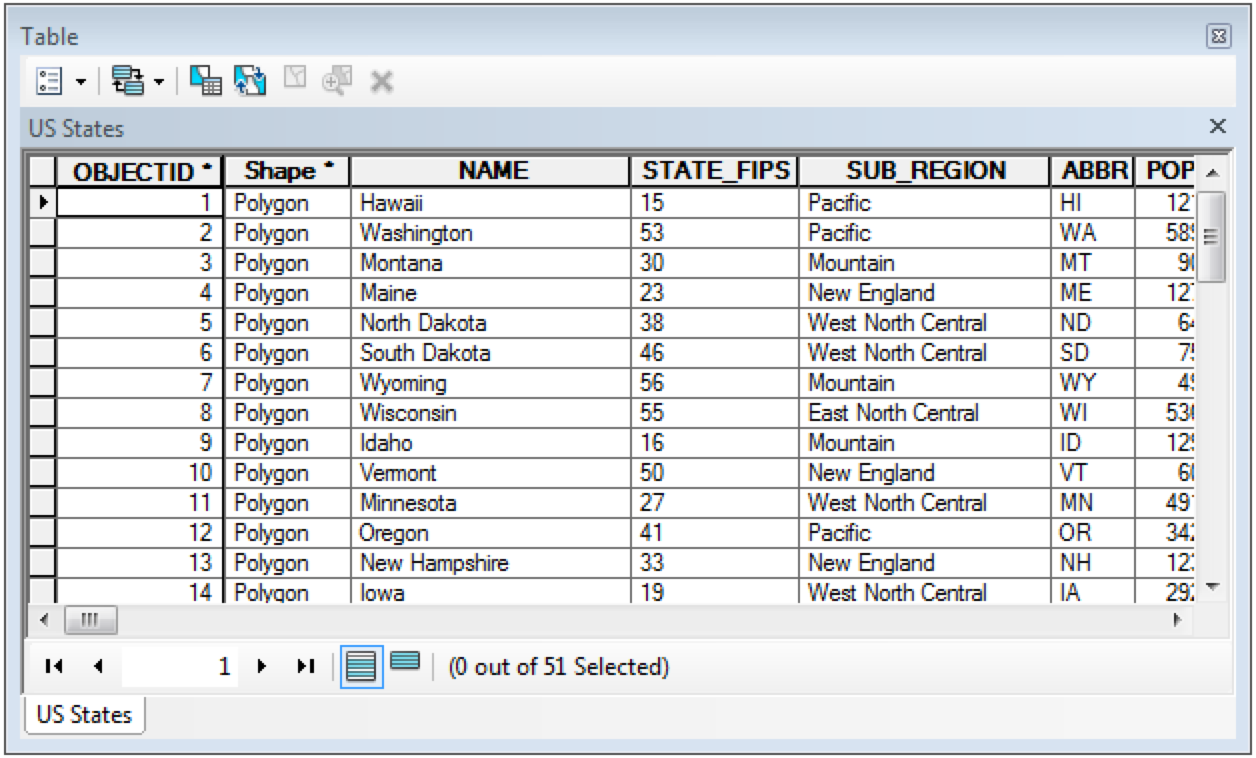
Time to think

- What kind of storage would be used to store the following kinds of data? How many bytes are needed for e ld?
  - Last name
  - Number of children in househ
  - Student grade point average
  - Year of birth
  - County population
  - Aerial image of county

Field Type
Short
Long
Float
Double
Text
Date
BLOB*

Tables in ArcGIS

- Tables contain attribute data
- Many formats, one interface



Table

US States

OBJECTID *	Shape *	NAME	STATE_FIPS	SUB_REGION	ABBR	POP
1	Polygon	Hawaii	15	Pacific	HI	12
2	Polygon	Washington	53	Pacific	WA	58
3	Polygon	Montana	30	Mountain	MT	9
4	Polygon	Maine	23	New England	ME	12
5	Polygon	North Dakota	38	West North Central	ND	6
6	Polygon	South Dakota	46	West North Central	SD	7
7	Polygon	Wyoming	56	Mountain	WY	4
8	Polygon	Wisconsin	55	East North Central	WI	53
9	Polygon	Idaho	16	Mountain	ID	12
10	Polygon	Vermont	50	New England	VT	6
11	Polygon	Minnesota	27	West North Central	MN	49
12	Polygon	Oregon	41	Pacific	OR	34
13	Polygon	New Hampshire	33	New England	NH	12
14	Polygon	Iowa	19	West North Central	IA	29

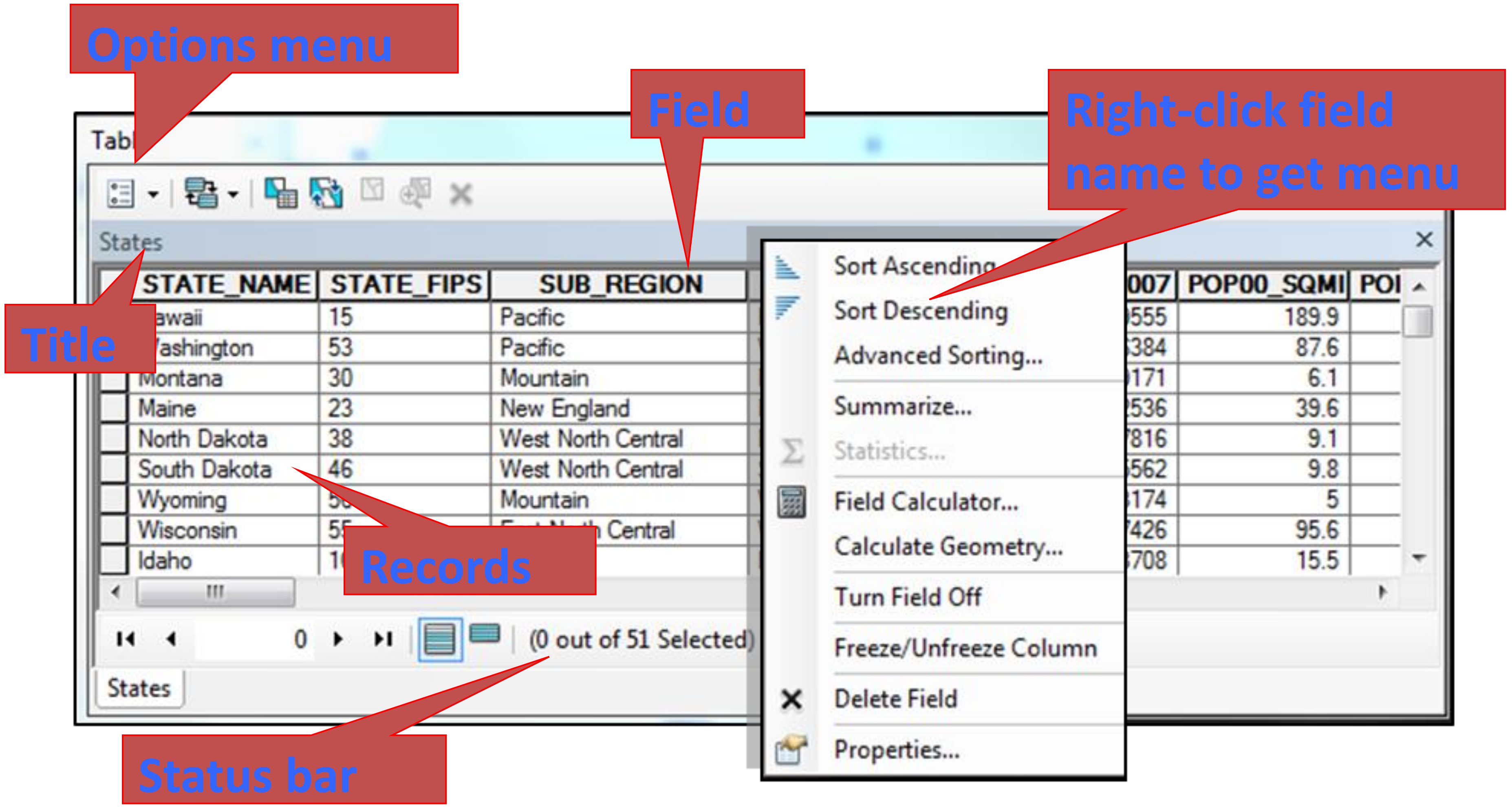
1 (0 out of 51 Selected)

US States

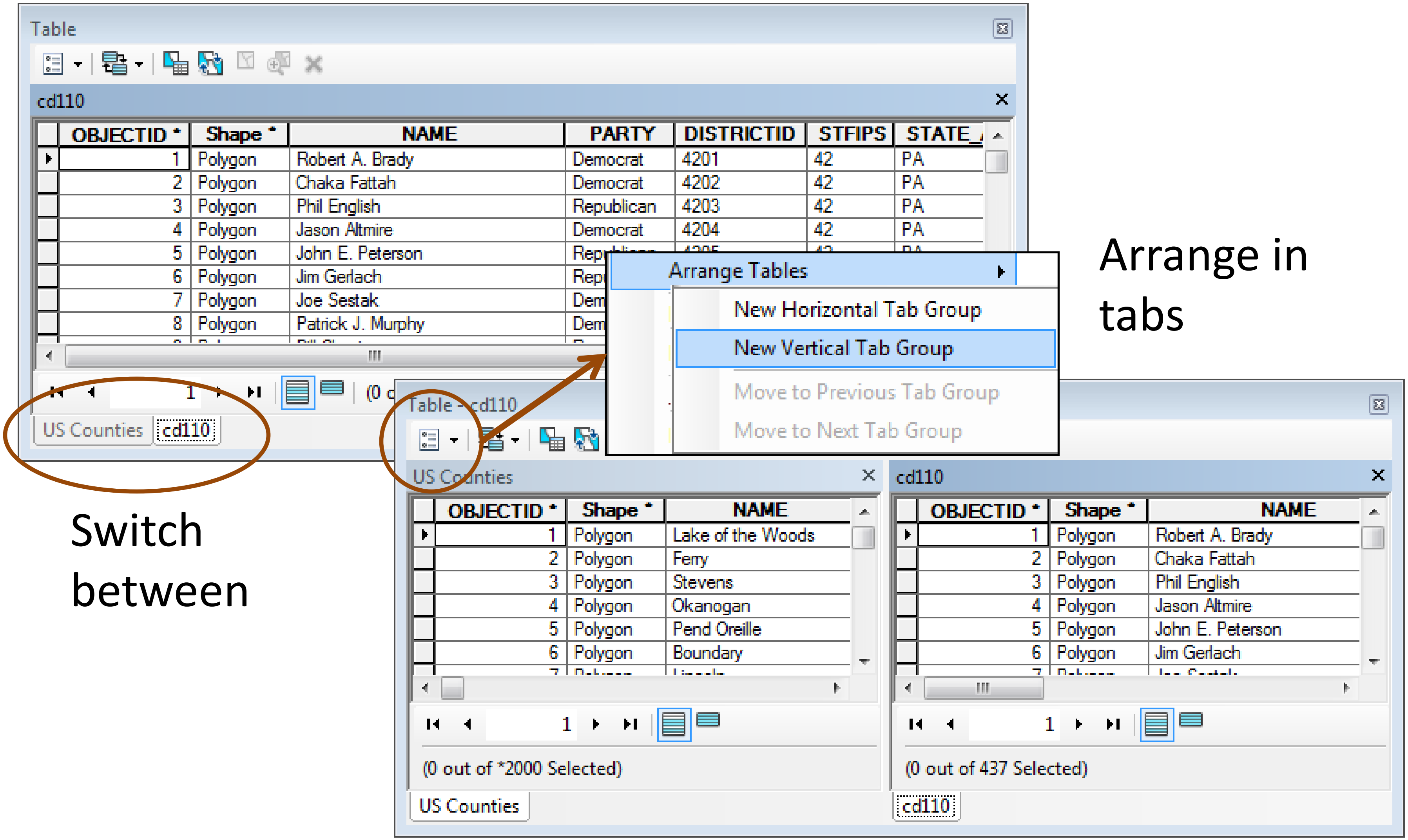
Sources of tables

- Dbase files
- INFO files
- ASCII Text files (tab or comma delimited)
- Records from SQL database systems
- Excel worksheets

ArcMap table interface



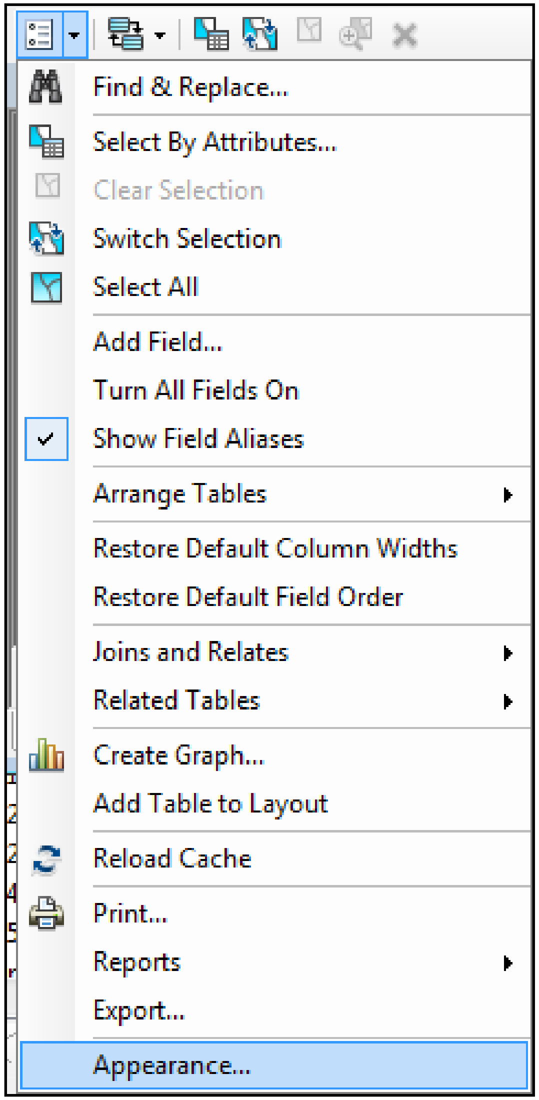
Managing multiple tables

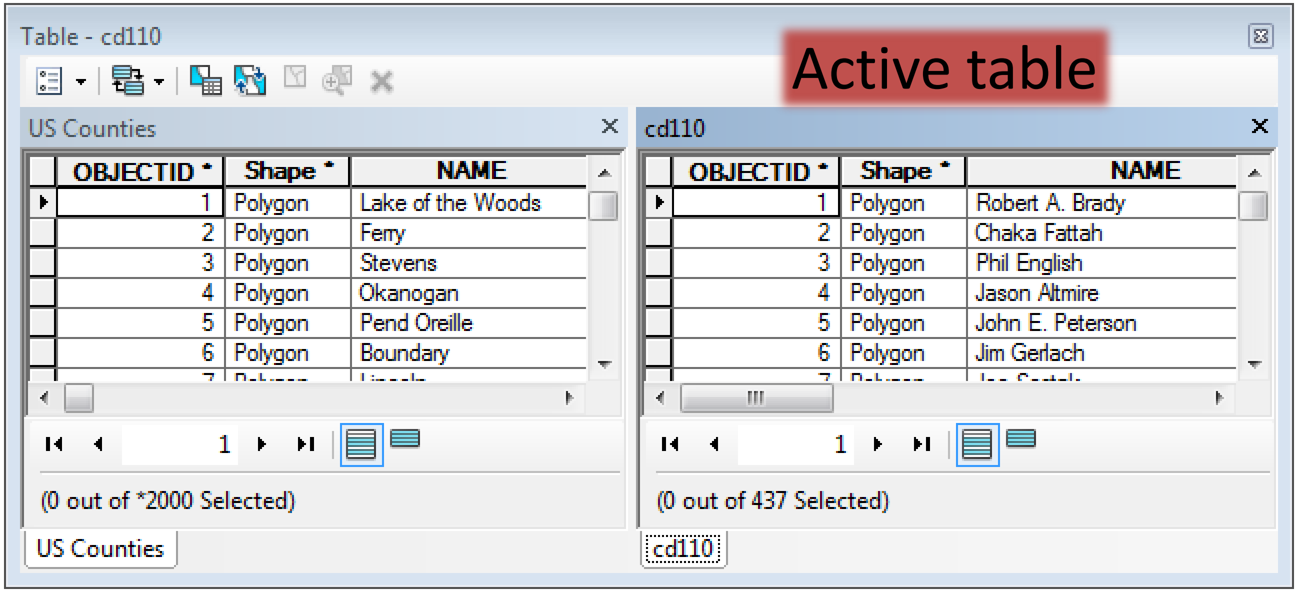


Arrange in tabs

Switch between

Table Options menu

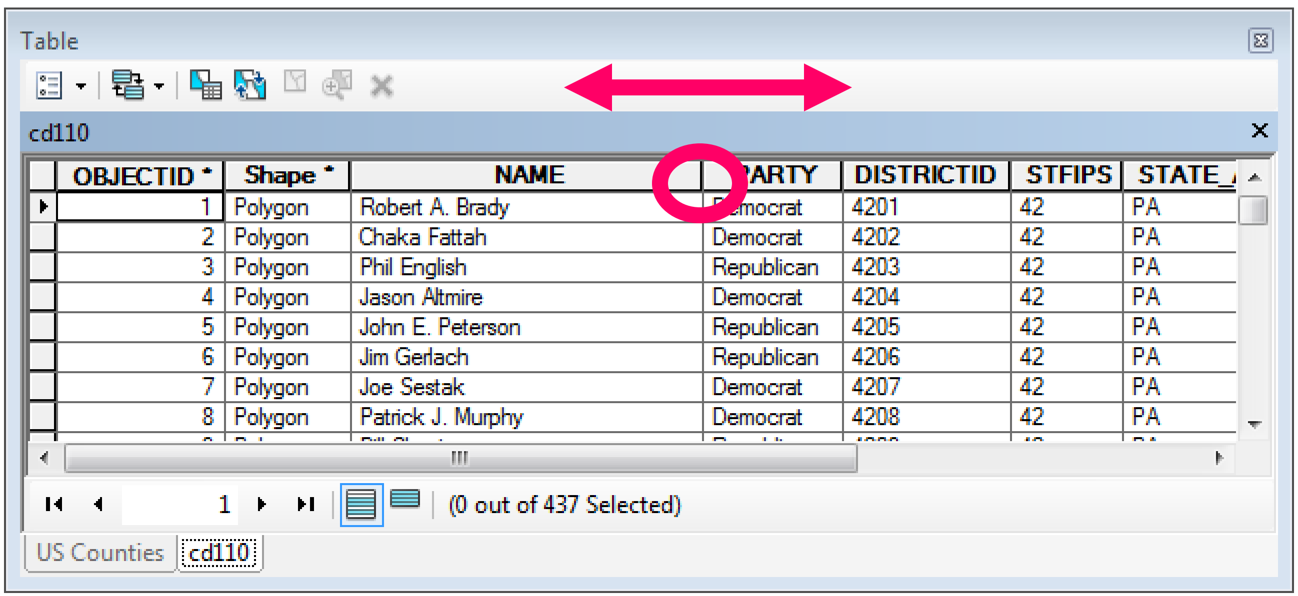




Commands affect the currently active table

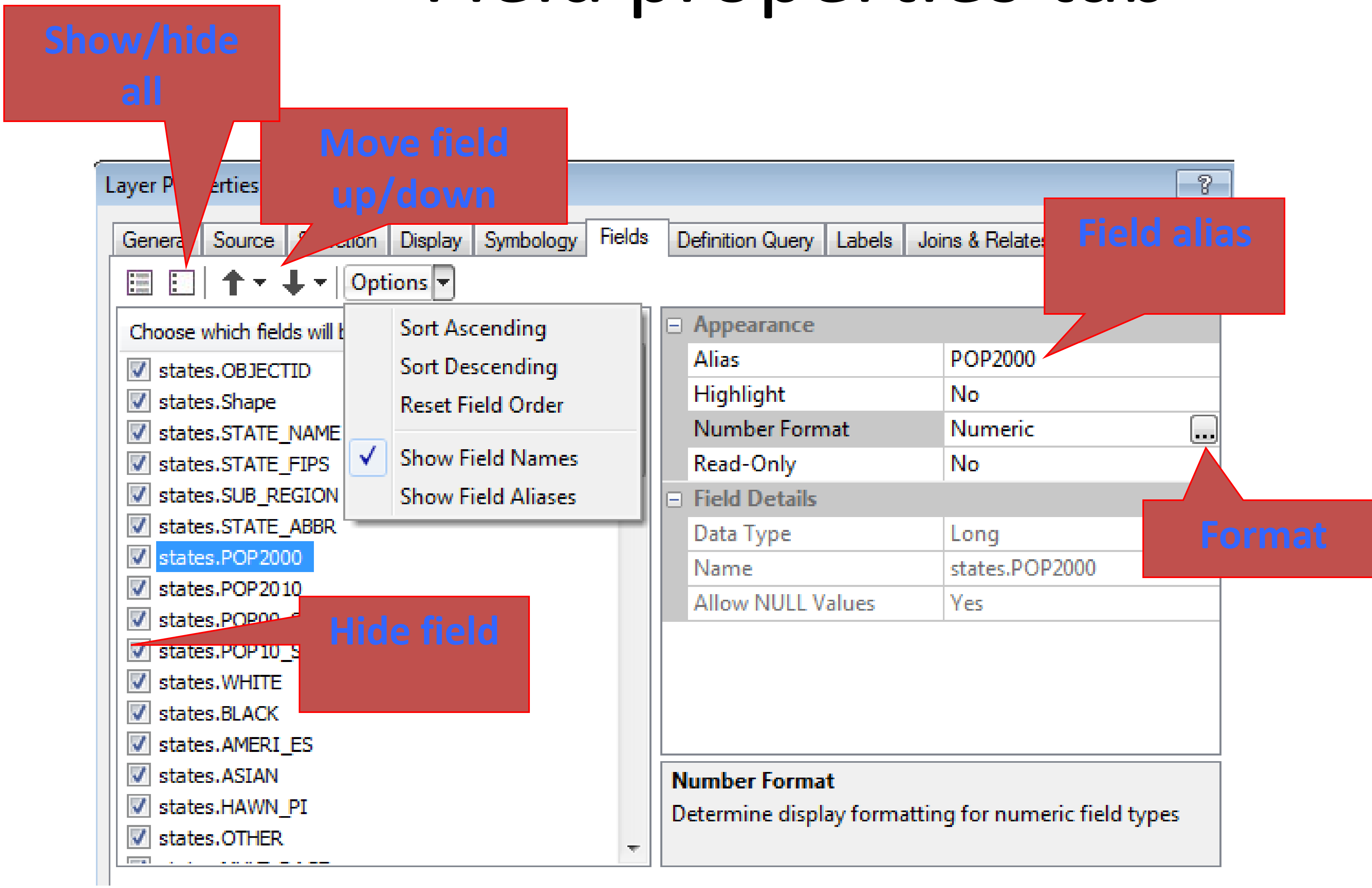
Adjusting field width

- Temporary, does not affect stored file

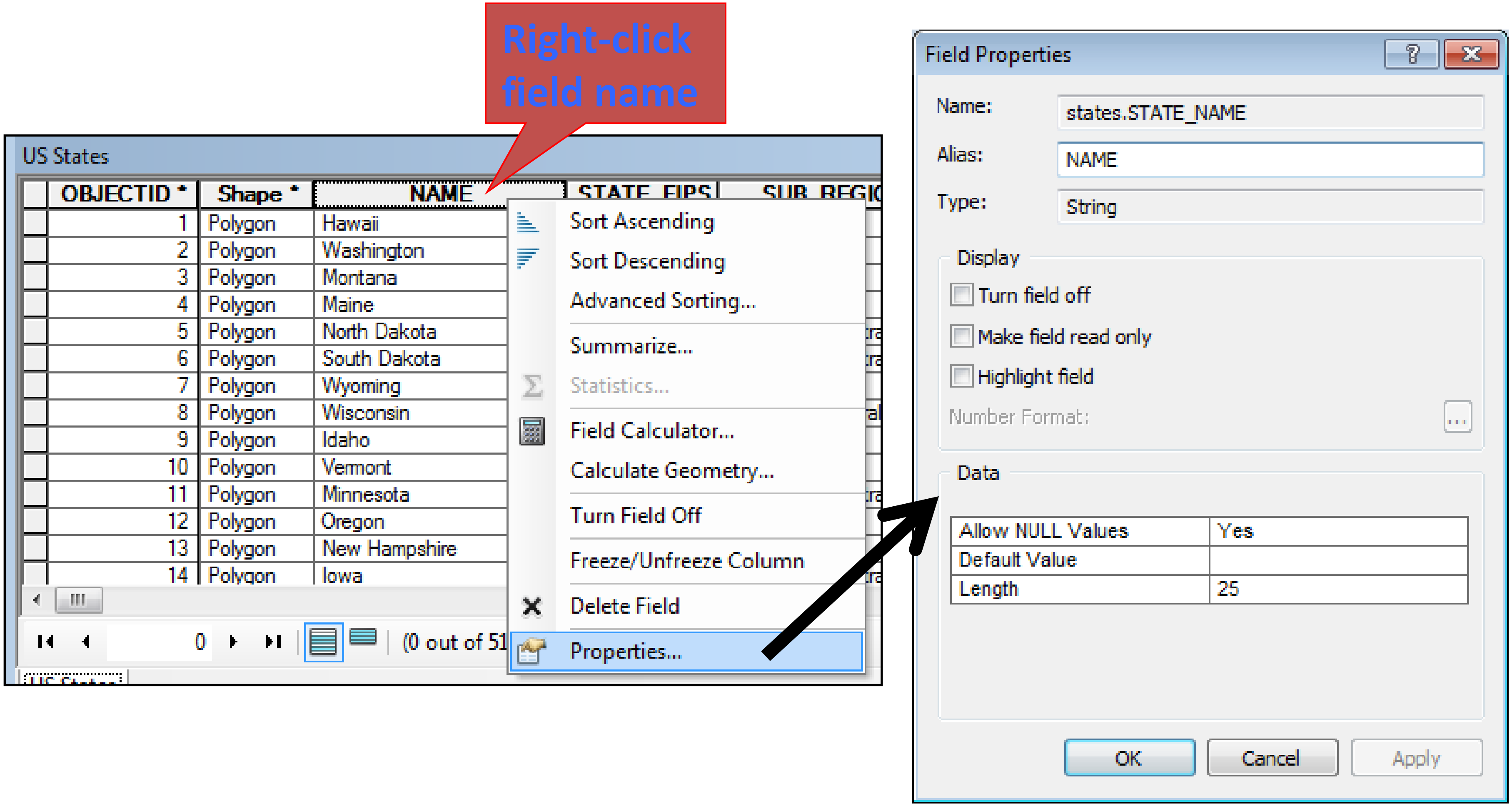


Hover over field break to get double arrow, then drag

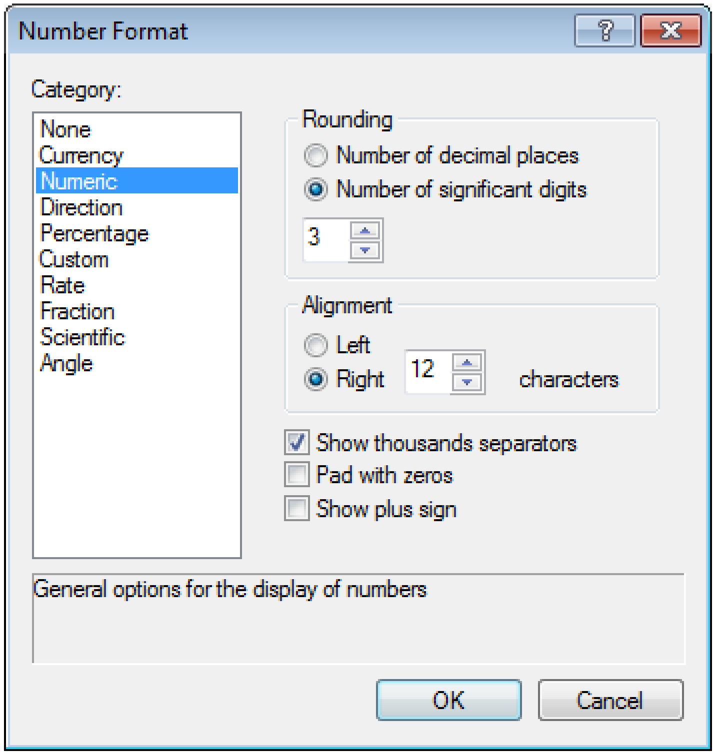
Field properties tab



Shortcut to field properties



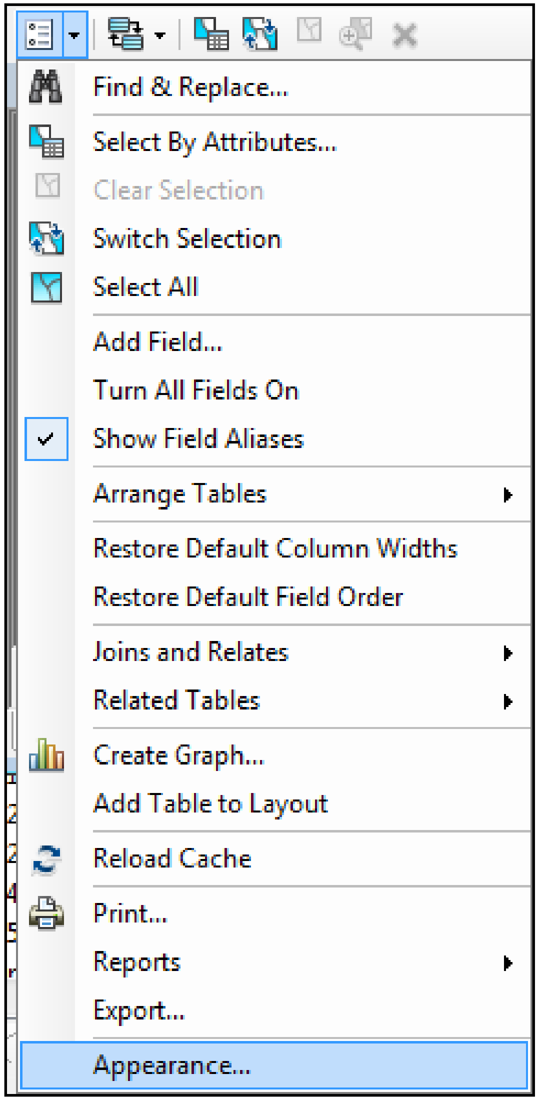
Formatting field display

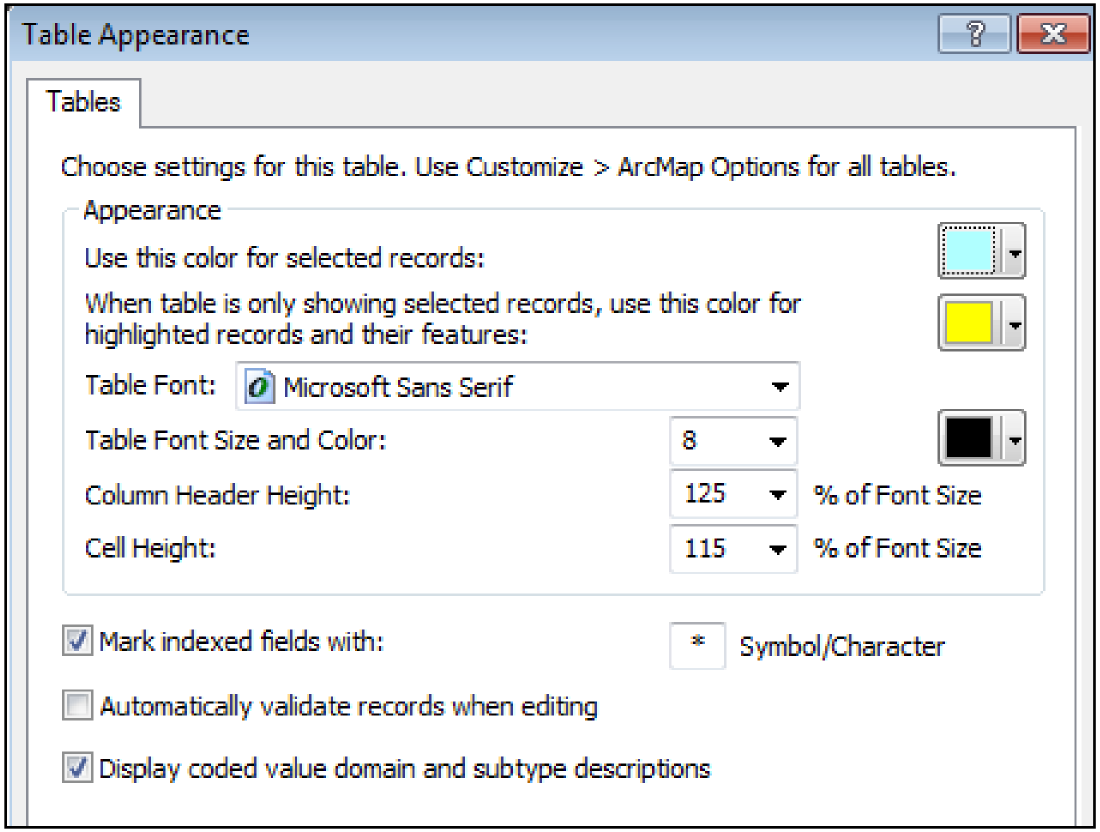


POP2010
1309580
6756150
983932
1338645
662194
827263
548154

POP2010
1,310,000
6,760,000
984,000
1,340,000
662,000
827,000
548,000

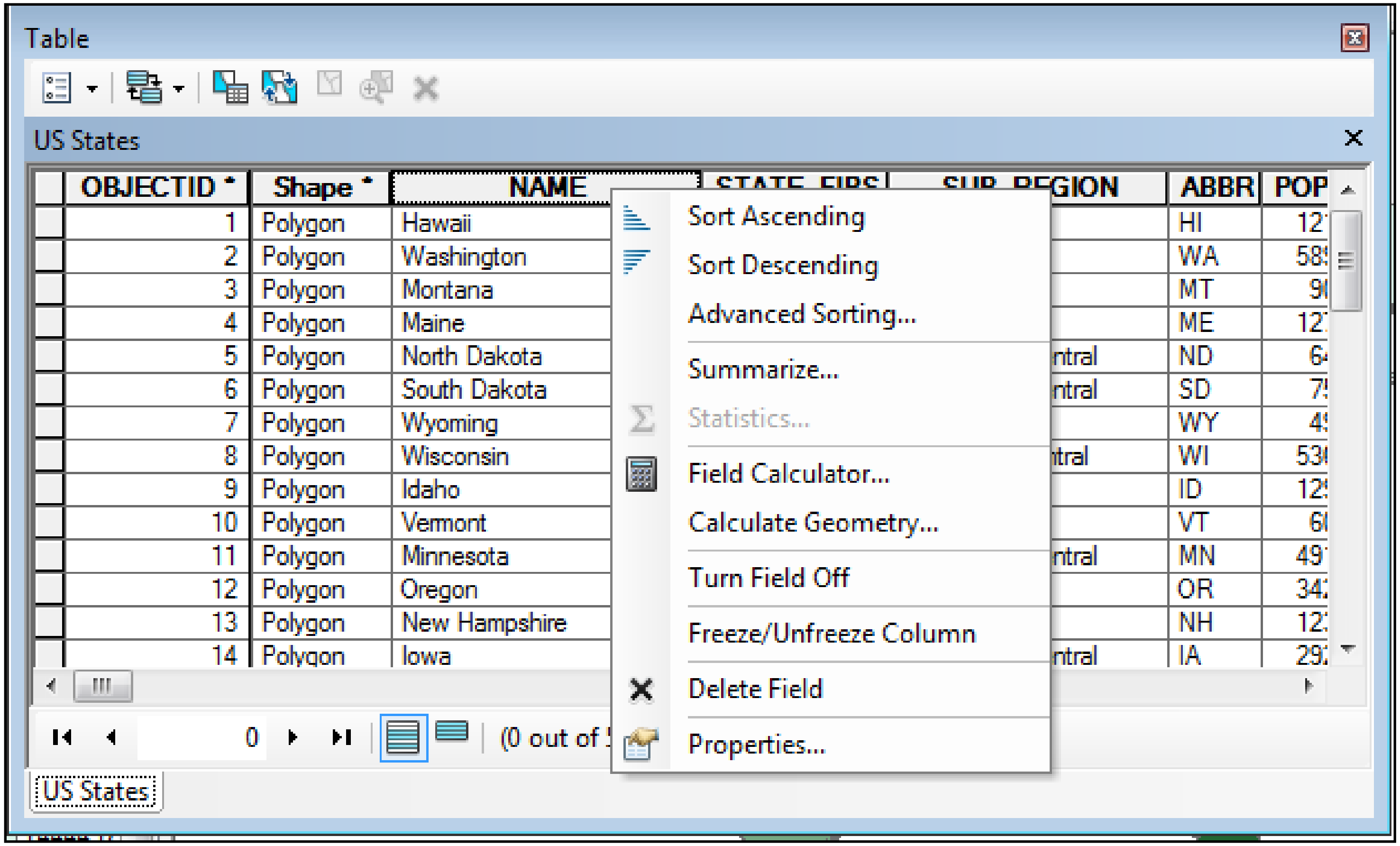
Table appearance



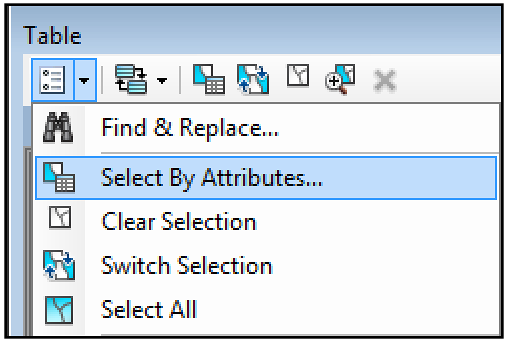


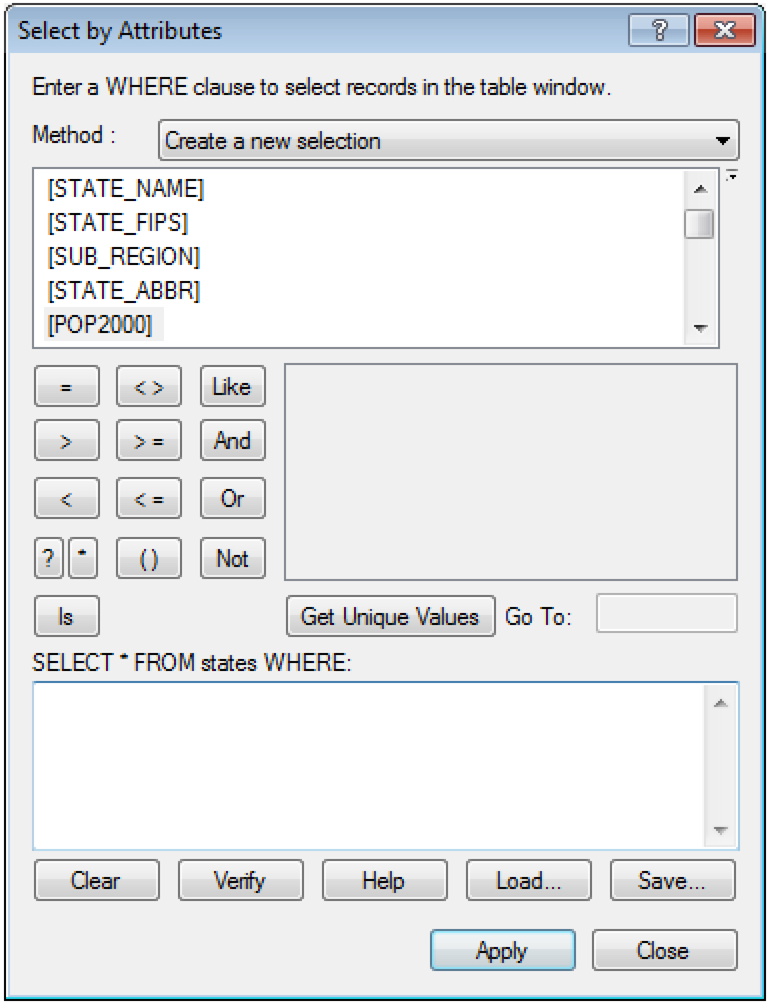
Sorting tables

- Has no effect on original data



Select by Attributes





Some valid queries

[POP1990] > 1000000

"STATE_NAME" = 'Alabama'

[POP2000] >= [POP1990]

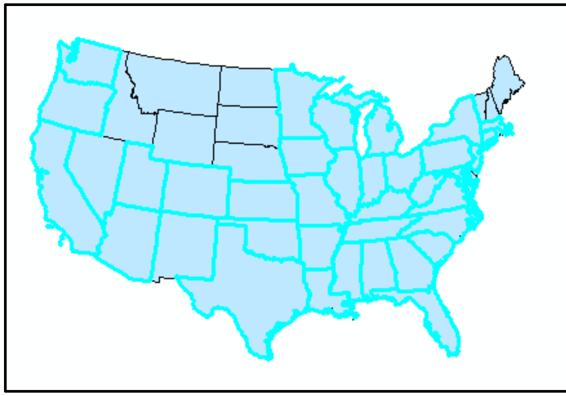
DBF tables have field names enclosed in quotes

Geodatabase tables have field names enclosed in brackets

Show selected

SELECT * FROM states WHERE:

"POP2000" > 2000000



Table

US States

OBJ	Shape *	NAME	ST	SUB_REGION	ABBR	POP2000	POP2010
7	Polygon	Wyoming	56	Mountain	WY	493782	548,000
8	Polygon	Wisconsin	55	East North Central	WI	5363675	5,740,000
9	Polygon	Idaho	16	Mountain			
10	Polygon	Vermont	50	New Englar			
11	Polygon	Minnesota	27	West North			
12	Polygon	Oregon	41	Pacific			
13	Polygon	New Hampshire	33	New Englar			
14	Polygon	Iowa	19	West North			
15	Polygon	Massachusetts	25	New Englar			
16	Polygon	Nebraska					

0 (34 out of 51 Sele

US States

Table

US States

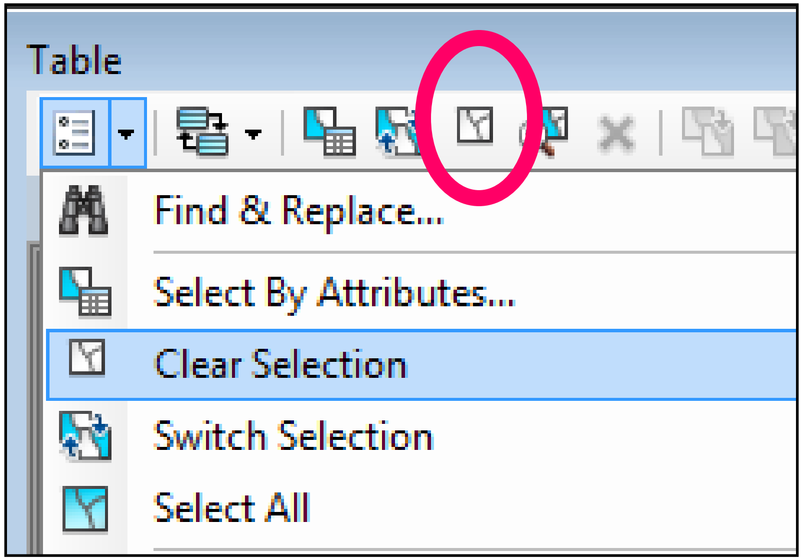
OBJ	Shape *	NAME	ST	SUB_REGION	ABBR	POP2000	POP2010
2	Polygon	Washington	53	Pacific	WA	5894121	6,760,000
8	Polygon	Wisconsin	55	East North Central	WI	5363675	5,740,000
11	Polygon	Minnesota	27	West North Central	MN	4919479	5,330,000
12	Polygon	Oregon	41	Pacific	OR	3421399	3,870,000
14	Polygon	Iowa	19	West North Central	IA	2926324	3,060,000
15	Polygon	Massachusetts	25	New England	MA	6349097	6,560,000
17	Polygon	New York	36	Middle Atlantic	NY	18976457	19,500,000
18	Polygon	Pennsylvania	42	Middle Atlantic	PA	12281054	12,600,000
19	Polygon	Connecticut	09	New England	CT	3405565	3,540,000
21	Polygon	New Jersey	34	Middle Atlantic	NJ	8414350	8,820,000

(34 out of 51 Selected)

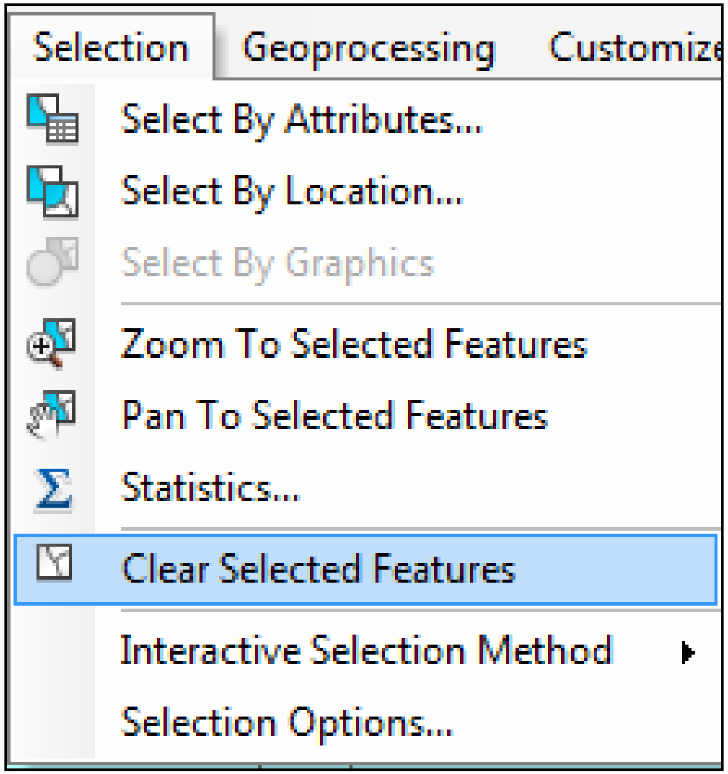
US States

Clear Selection

In Table window



From main menu

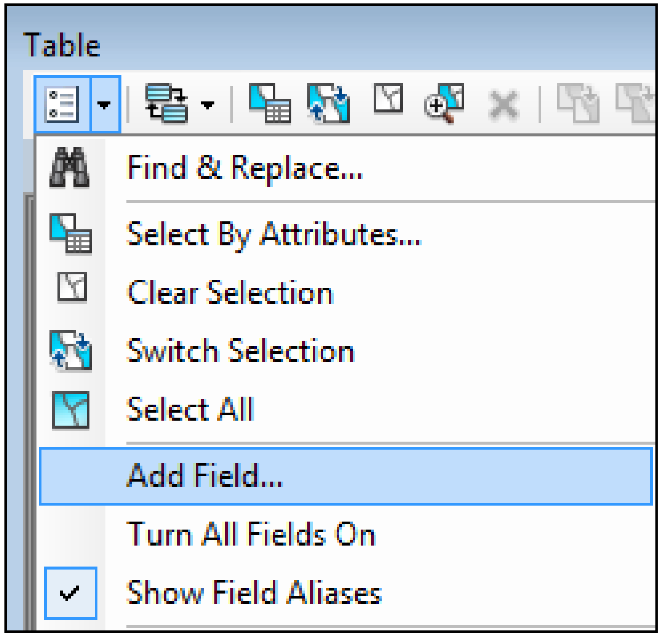


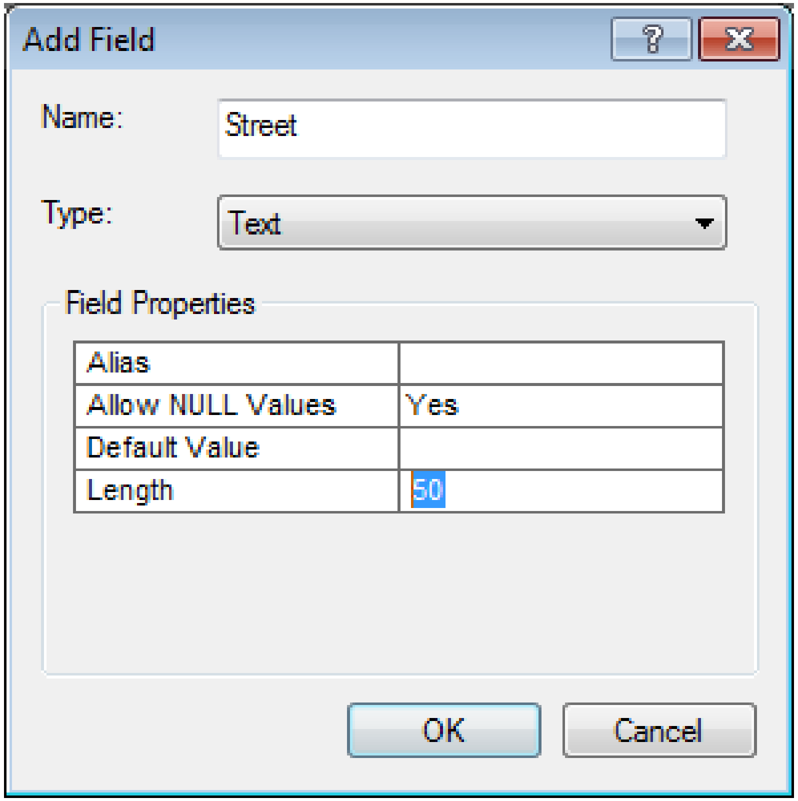
ArcGIS field data types

Field Type	Explanation	Examples
Short	Integers stored as 2-byte binary numbers *Range of values −32,000 to +32,000*	255 12001
Long	Integers stored as 10-byte binary numbers *Range of values −2.14 billion to +2.14 billion*	156000 457890
Float	Floating-point values with eight significant digits in the mantissa	1.289385e12 1.5647894e–02
Double	Double-precision floating-point values with 16 significant digits in the mantissa	1.12114118119141e13
Text	Alphanumeric strings	'Maple St' 'John H. Smith'
Date	Date format	07/12/92 10/17/63
BLOB*	Binary large object; any complex binary data including images, documents, etc.	

Adding a field

- In ArcMap

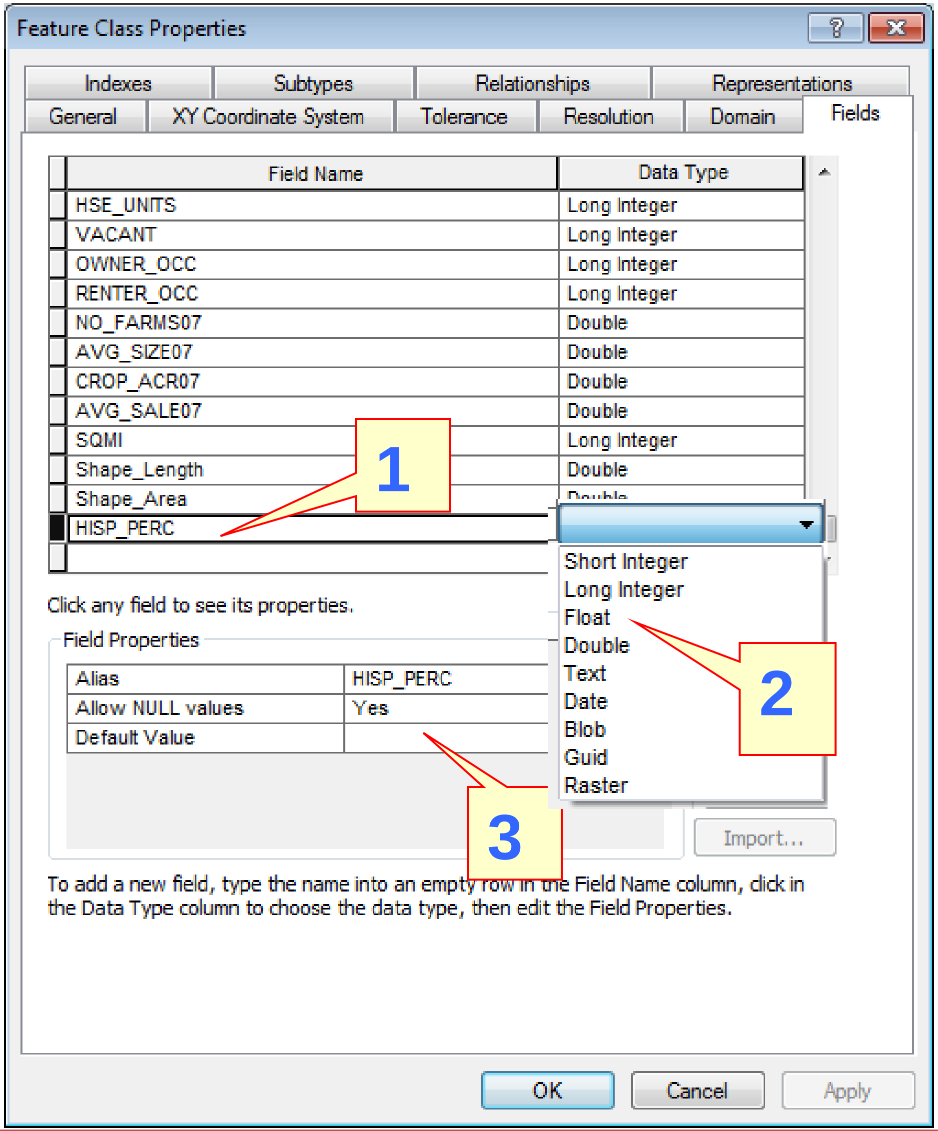




Think about the field type before you add the field!

Adding multiple fields

- Easier in the Catalog properties
  - Type name in empty row
  - Set field type
  - Set field properties
  - Click Apply
  - Add next…

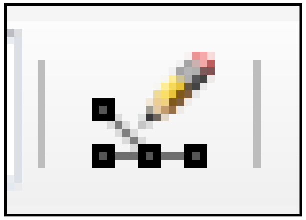


Think about the field type before you add the field!

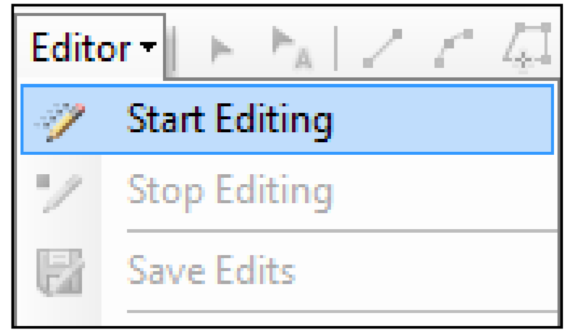
Editing and calculating fields

Editing fields

Open Editor toolbar



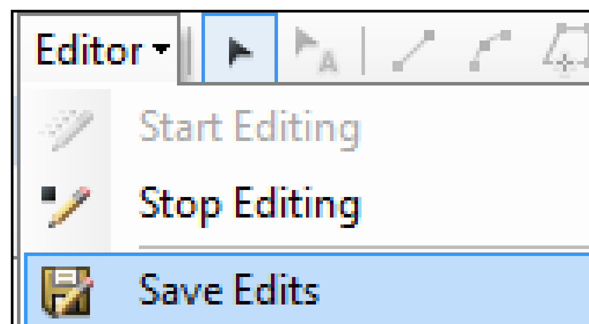
Start editing



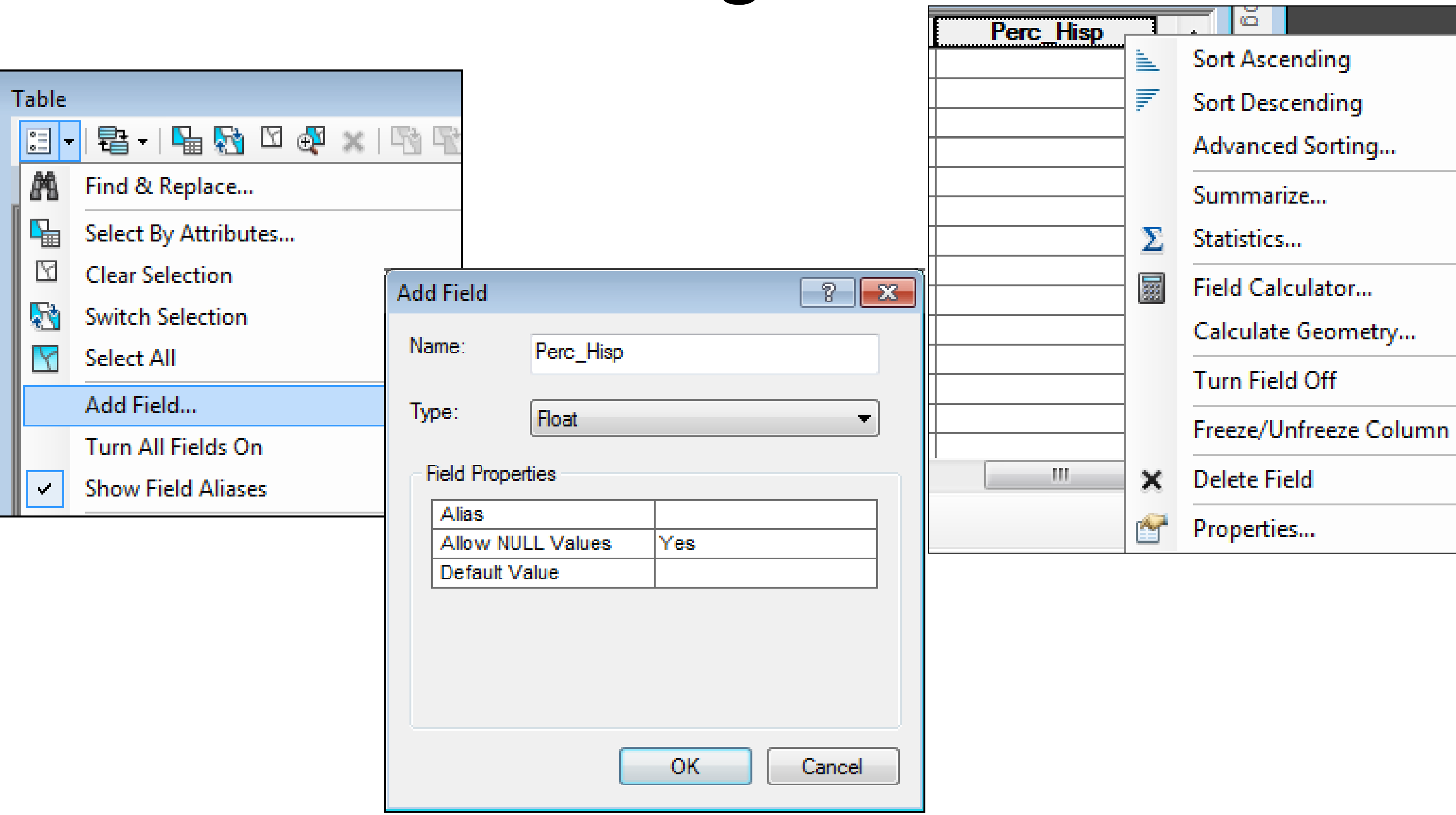
Type edits in fields

Ave_MAG	Ave_MMI	Risk
6.5042	3.7358	High
3.6143	6.1429	Low
5.0113	6.5625	High
2.135	5.9	Lo
6.35	7	
6.3167	8.5	

Save edits, stop editing



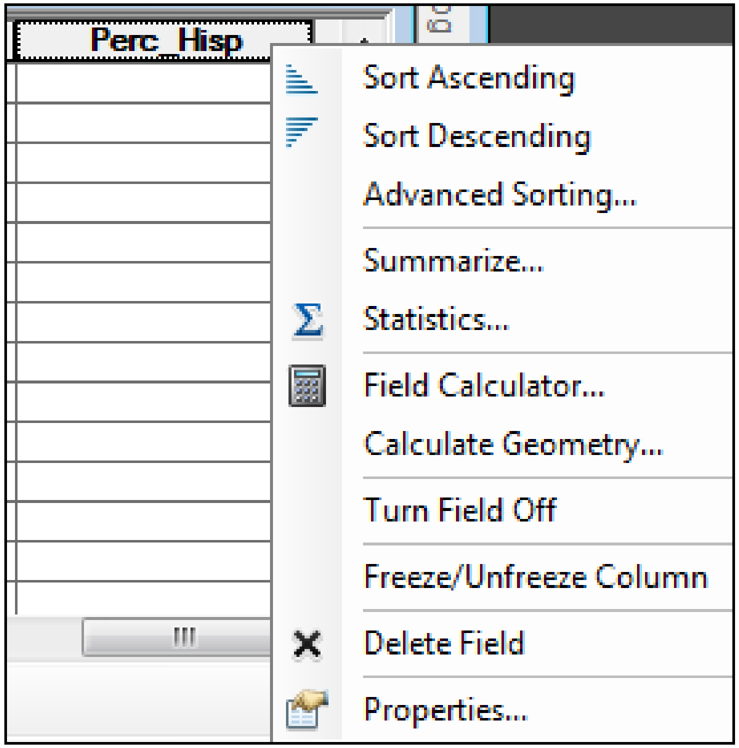
Calculating fields



Add a new field if necessary

Consider whether you need decimal places!

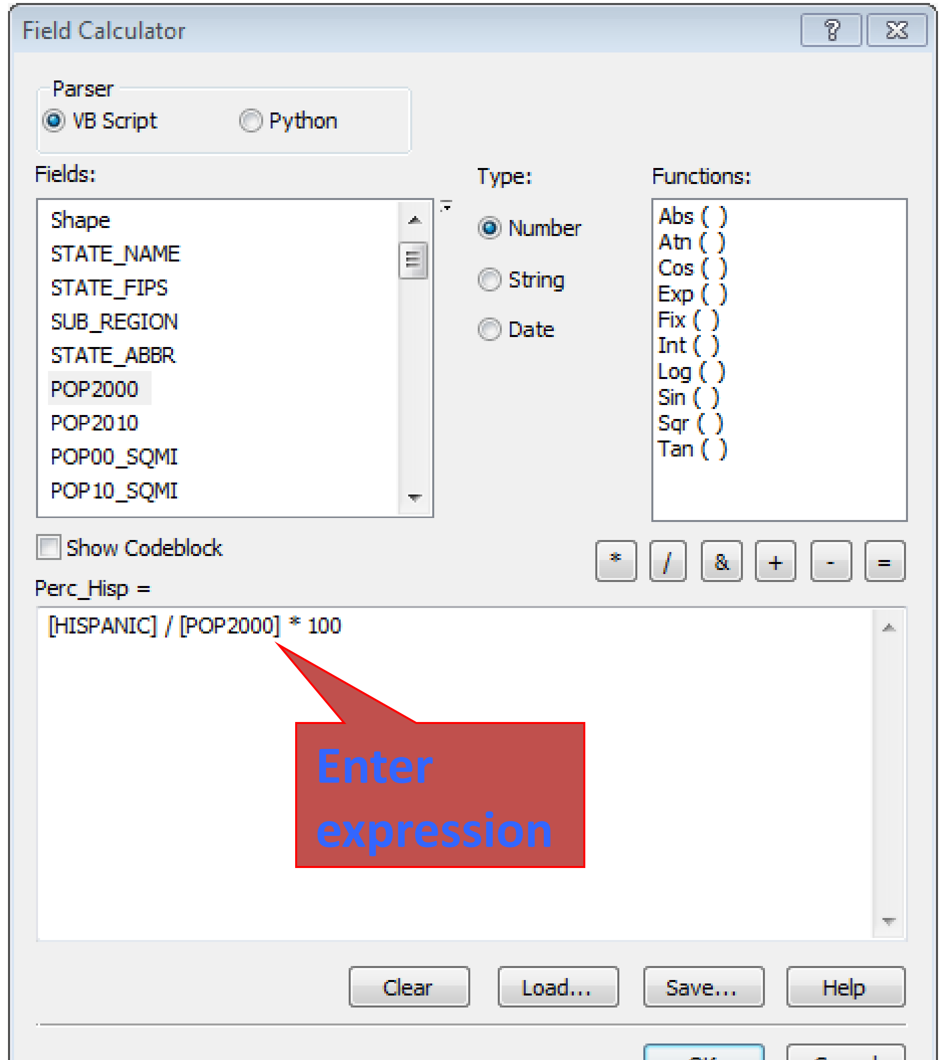
Calculate



Right-click field to calculate

Usually field should be EMPTY

Be careful! Calculations can't be undone unless you are in an edit session



Geodatabase geometry measures

- Automatically created and maintained
  - Usually appear at end of table
    - Shape_Area field
    - Shape_Length field
  - Units will match units of the coordinate system

US States

AVG_SIZE07	CROP_ACR0	AVG_SALE07	SQMI	Shape_Length	Shape_Area
149	177626	68.29	10931	1275835.715167	16513001143.5337
381	7609210	172.92	71297	3004175.455006	164442316568.045
2079	18241710	94.94	14703	2859079.544323	359662292212.805
166	529253	75.86	39555	1804888.364454	78384814218.1752
1241	27527180	190.31	70698	1753454.614357	173072126682.401
1401	19094311	210.8	77115	1943874.72695	187970221480.16
2726	2576017	104.58	97813	1960001.350821	238004239410.541
194	10116279	114.29	59822	1968992.210358	136605586997.204

0 (0 out of 51 Selected)

Shapefile geometry measures

- Shapefiles DO NOT create or maintain area/length fields automatically!
  - Must be created and updated manually
  - Some functions and operations can change the lengths and areas of features
  - If you find an AREA field there is no guarantee that it is correct

watersheds

	FID	Shape *	AREA	PERIMETER	NAME	AREA_ACRE_
	0	Polygon	5265900	16620	Nameless Cave	1301.23
	1	Polygon	765900	5700	Robbinsdale Park East	189.257
	2	Polygon	1311120	27480	Red Rock Canyon	3239.8401
	3	Polygon	2283300	9780	Wonderland Drive	564.21301
	4	Polygon	511200	4140	Highway 79 Industrial Park	126.32
	5	Polygon	4786200	15780	South Robbinsdale	1182.6899
	6	Polygon	1192060	25782.6	Landfill	2945.6299

Be careful!

Table

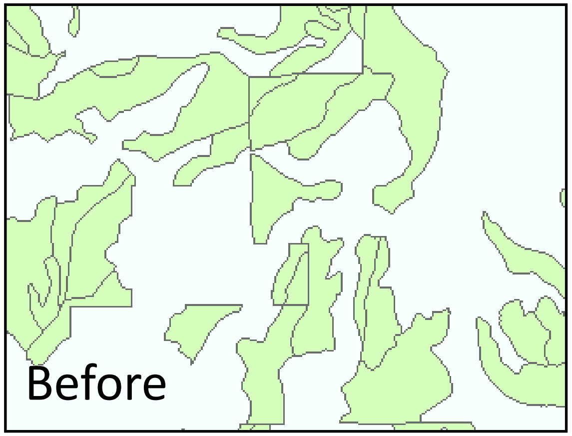
watersheds

FID	Shape *	AREA	PERIMETER	NAME	AREA_ACRE_	PERCENT	F
0	Polygon	5265900	16620	Nameless Cave	1301.23	3.0465	
1	Polygon	765900	5700	Robbinsdale Park East	189.257	0.443099	
2	Polygon	1311120	27480	Red Rock Canyon	3239.8401	7.58528	
3	Polygon	2283300	9780	Wonderland Drive	564.21301	1.32097	
4	Polygon	511200	4140	Highway 79 Industrial Park	126.32	0.295747	
5	Polygon	4786200	15780	South Robbinsdale	1182.6899	2.76898	
6	Polygon	1192060	25782.6	Landfill	2945.6299	6.89646	
7	Polygon	1014020	22008.6	Truck By-Pass	2505.6899	5.86647	

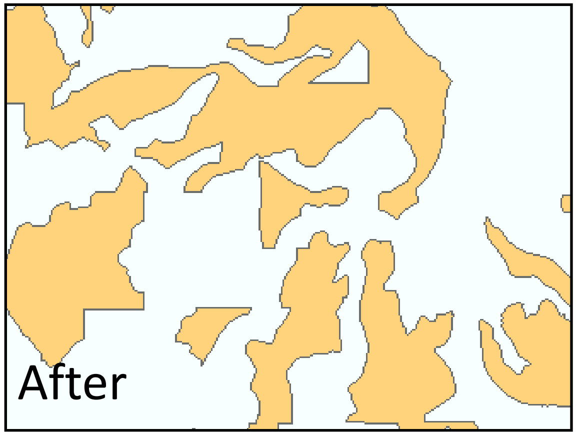
0 (0 out of 26 Selected)

watersheds

AREA/LENGTH/PERIMETER fields in shapefiles are NOT updated when features change.



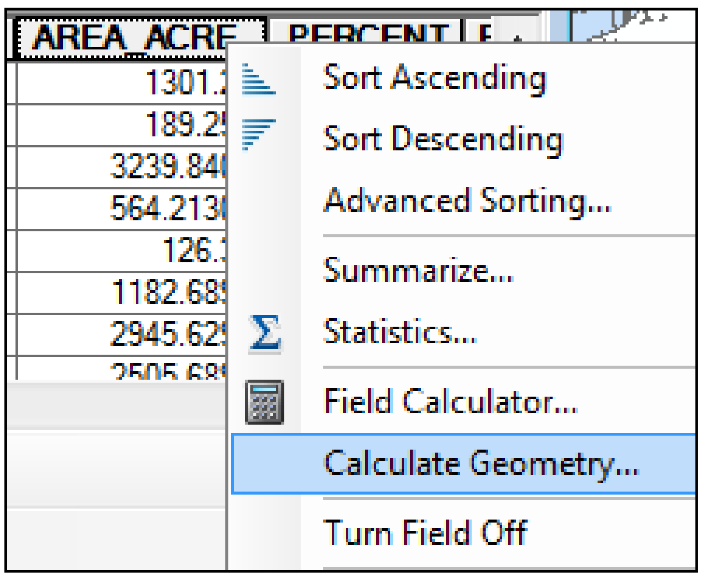
Before



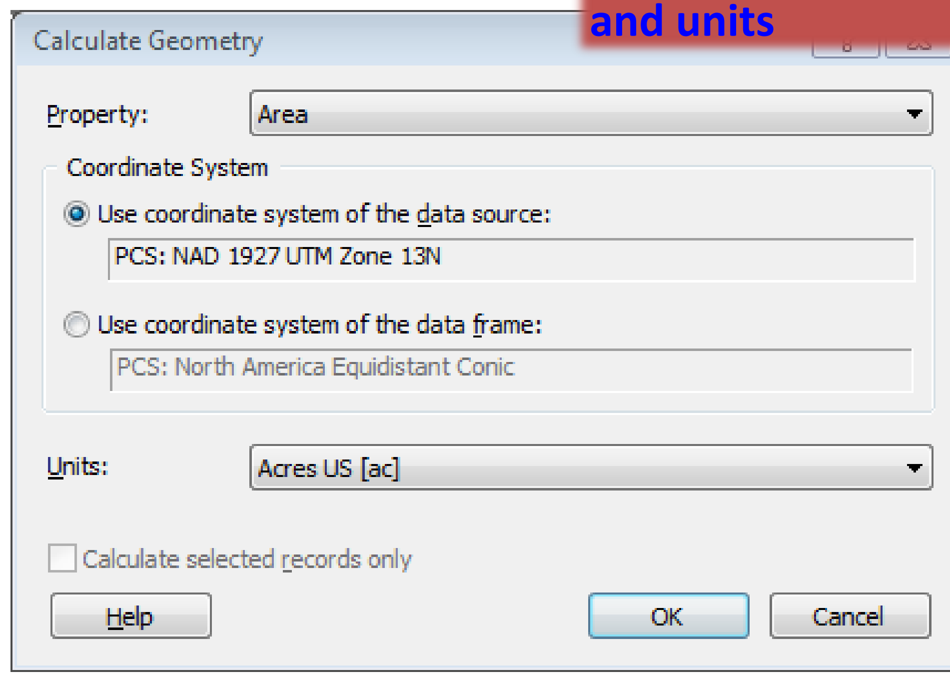
After

User geometry calculations

Right-click empty or incorrect field to update



Choose type, coordinate system, and units

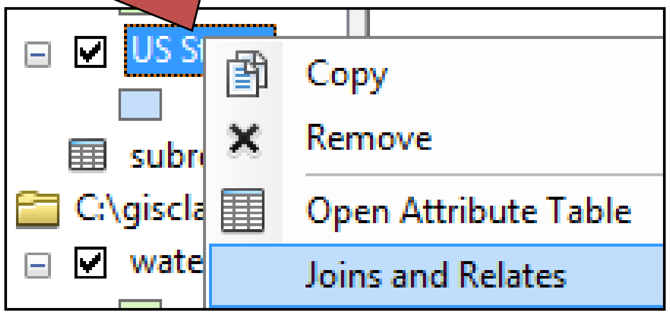


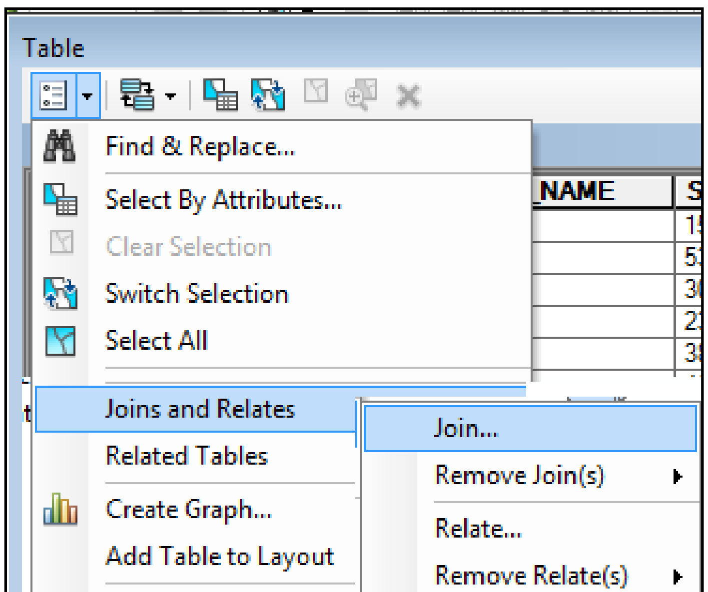
Test yourself: True or False?

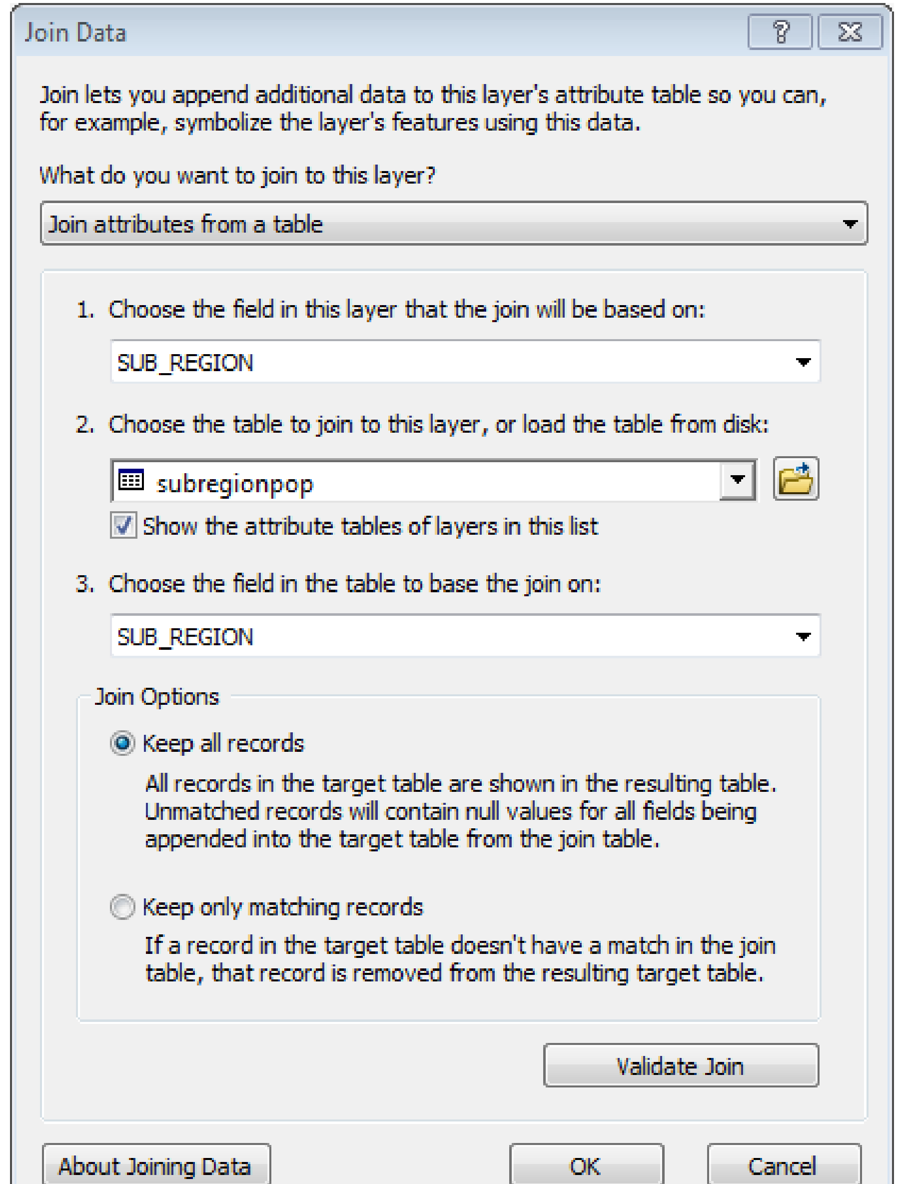
- A field named AREA in a shapefile will always have the correct area?
  - **False**. Shapefile area fields are not automatically maintained.
- A field named AREA in a geodatabase will always have the correct area?
  - **False**. Automatically updated fields in a geodatabase are called Shape_Area.

How to join or relate tables

Right-click destination in the Table of Contents







Importing tables

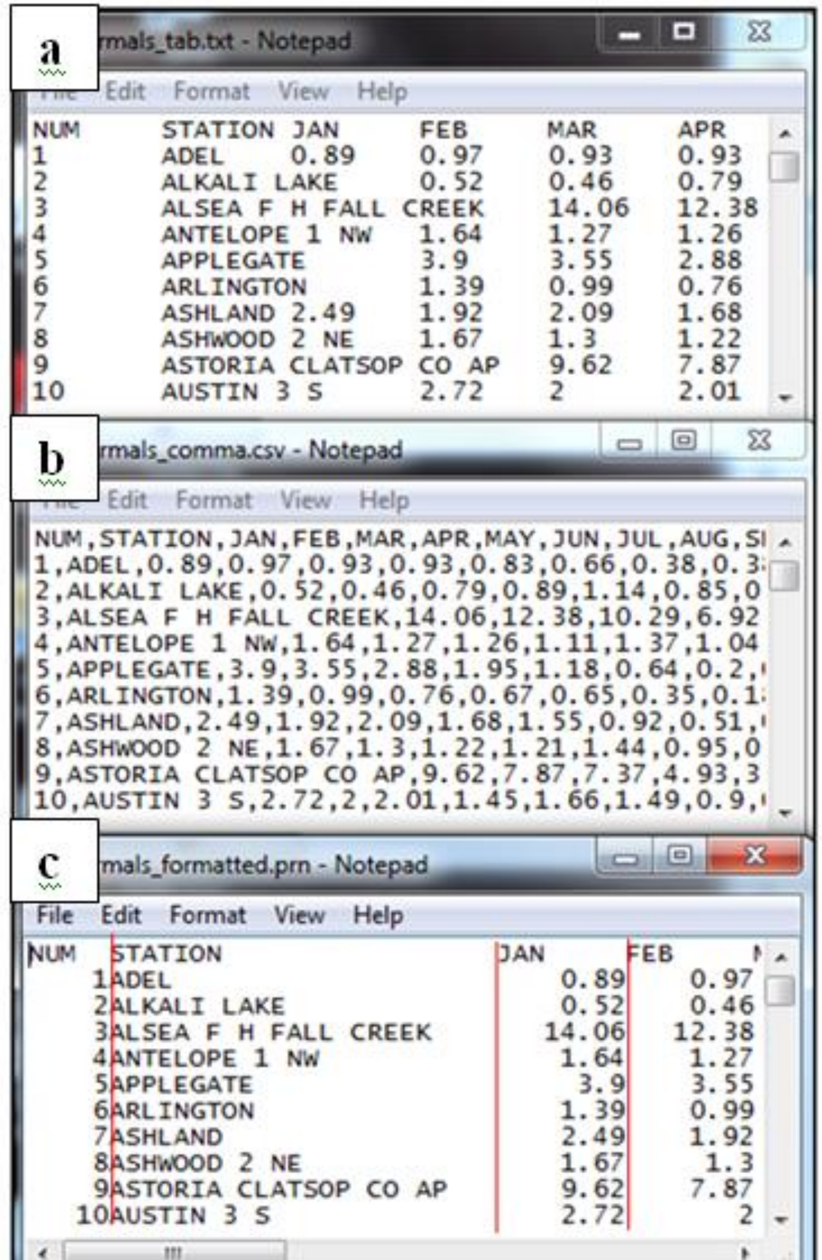
Allowed in ArcGIS

a) Tab-delimited

b) Comma-delimited

Not allowed!

c) Fixed-column



Excel files

- Excel worksheets with suitable layout can be opened as tables in ArcGIS
- Most functions that do not involve changing the file will work (sort, query)
- Tables cannot be changed or edited

Excel file requirements

- First row must contain field headings with legal field names as defined earlier
- No blank rows or formulas should be used
- Each column should contain only text or only numbers.
- It is helpful for each column to be formatted as text, numeric, etc.
- ArcMap cannot open files that are already open in Excel—they must be closed first

Acceptable Worksheet

Columns formatted as text or numeric

Legal field names

ORstations.xls [Comp[illegible]y Mode]

	D	E	F	G	H	I	J	K
1	STATION NAME	LATDEG	LATMIN	LONDEG	LONMIN	ELEV	LAT	LON
2	ADEL	42	11	119	54	4583	42.183333	-119.900000
3	ALKALI LAKE	42	58	120	0	4332	42.966667	-120.000000
4	ALSEA F H FALL CREEK	44	24	123	45	230	44.400000	-123.750000
5	ANTELOPE 1 NW	44	55	120	44	2840	44.916667	-120.733333
6	APPLEGATE	42	15	123	10	1282	42.250000	-123.166667
7	ARLINGTON	45	43	120	12	277	45.716667	-120.200000
8	ASHLAND	42	13	122	43	1724	42.216667	-122.716667
9	ASHWOOD 2 NE	44	45	120	43	2820	44.750000	-120.716667
10	ASTORIA CLATSOP CO AP	46	9	123	53	9	46.150000	-123.883333
11	AUSTIN 3 S	44	34	118	29	4213	44.566667	-118.483333
12	BAKER	44	50	117	49	3368	44.833333	-117.816667
13	BANDON 2 NNE	43	9	124	24	20	43.150000	-124.400000
14	[illegible]	43	57	120	13	3970	43.950000	-120.216667
15	[illegible]	45	27	122	49	270	45.450000	-122.816667
16	[illegible] N	44	17	122	2	2152	44.283333	-122.033333
17	[illegible]	44	3	121	17	3660	44.050000	-121.283333

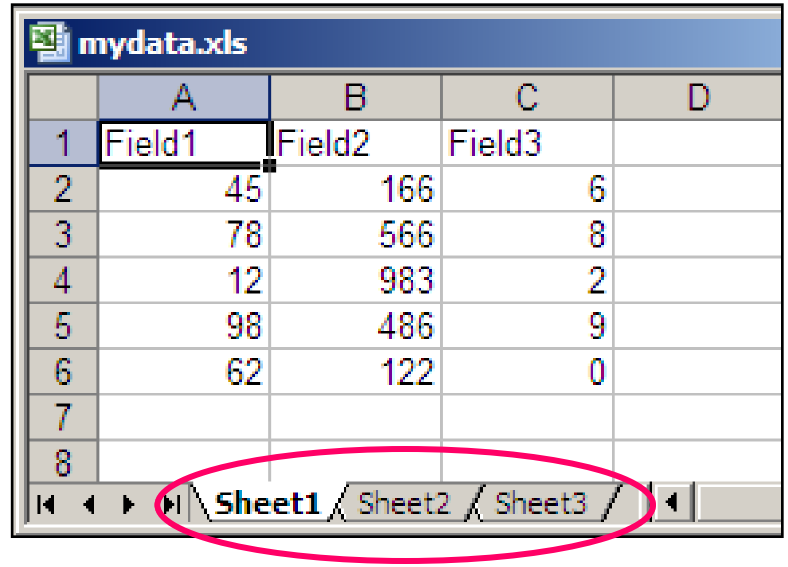
ORstations

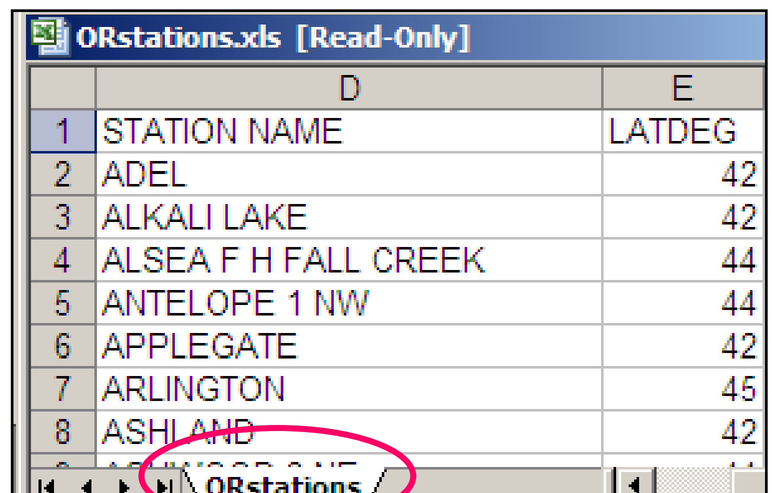
No formulas or blank lines

Decimal degrees = degrees + minutes/60 + seconds/3600

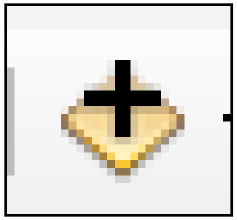
Workbooks and worksheets

- An Excel workbook file (.xls) may contain more than one worksheet.
- By default there are three named Sheet1, Sheet2, Sheet3.
- There may be one or more named worksheets.
- ArcMap can only open one worksheet at a time.
- You will open the workbook like a folder and select a single worksheet

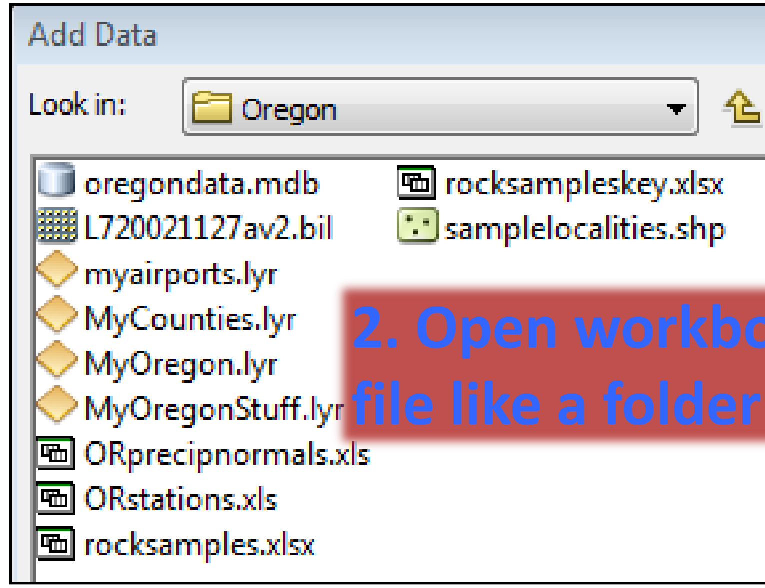




Adding a worksheet

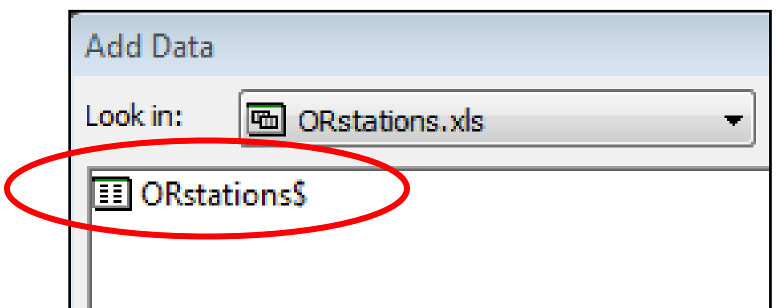


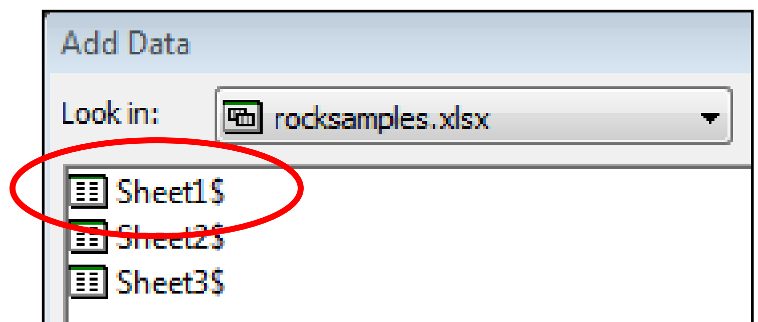
1. Add data button



2. Open workbook file like a folder

3. Add named worksheet, or the first unnamed worksheet.





Excel file in ArcMap

Table

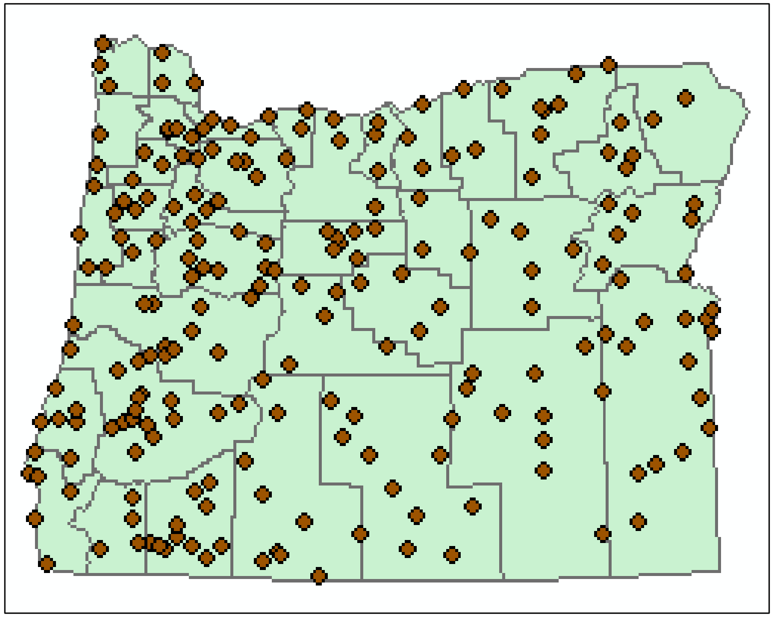
ORstations$

NUM	COOPID	TYPE	LATDEG	LATMIN	LONDE	LONMIN	ELEV	LAT	LON	STATION NAME
1	350036	XNP	42	11	119	54	4583	42.183333	-119.9	ADEL
2	350118	XNP	42	58	120	0	4332	42.966667	-120	ALKALI LAKE
3	350145	P	44	24	123	45	230	44.4	-123.75	ALSEA F H FALL CREEK
4	350197	XNP	44	55	120	44	2840	44.916667	-120.733333	ANTELOPE 1 NW
5	350217	P	42	15	123	10	1282	42.25	-123.166667	APPLEGATE
6	350265	XNP	45	43	120	12	277	45.716667	-120.2	ARLINGTON
7	350304	XNP	42	13	122	43	1724	42.216667	-122.716667	ASHLAND
8	350312	P	44	45	120	43	2820	44.75	-120.716667	ASHWOOD 2 NE
9	350328	XNP	46	9	123	53	9	46.15	-123.883333	ASTORIA CLATSOP CO AP
10	350356	XNP	44	34	118	29	4213	44.566667	-118.483333	AUSTIN 3 S
11	350412	XNP	44	50	117	49	3368	44.833333	-117.816667	BAKER
12	350471	XNP	43	9	124	24	20	43.15	-124.4	BANDON 2 NNE
13	350501	XNP	43	57	120	13	3970	[illegible]	[illegible]	BARNES STATION
14	350595	XNP	45	27	122	49	270	[illegible]	[illegible]	BEAVERTON 2 SSW
15	350652	XNP	44	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	BELKNAP SPRINGS 8 N
16	350694	XNP	44	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	BEND

You can use x-y locations in a table to produce points on a map.

But first you must know the coordinate system of these locations. What is this one?

Displaying x-y values from a table



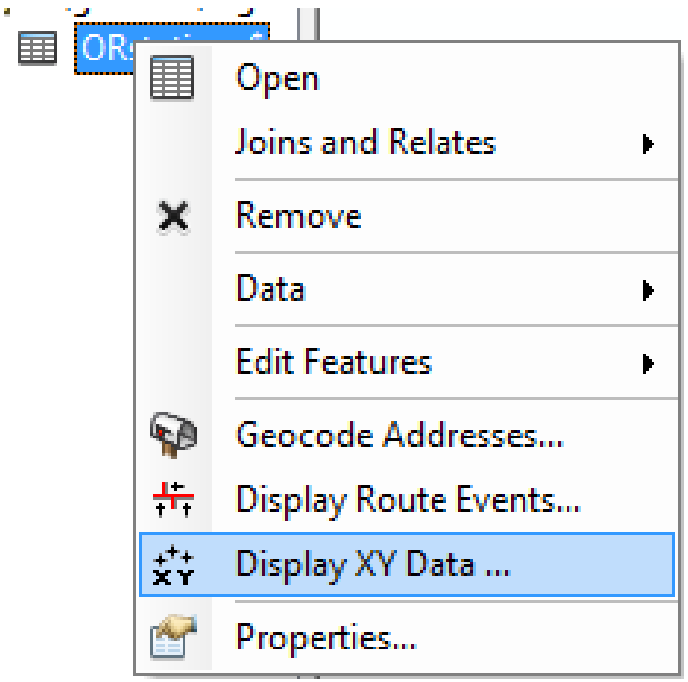
ORstations$

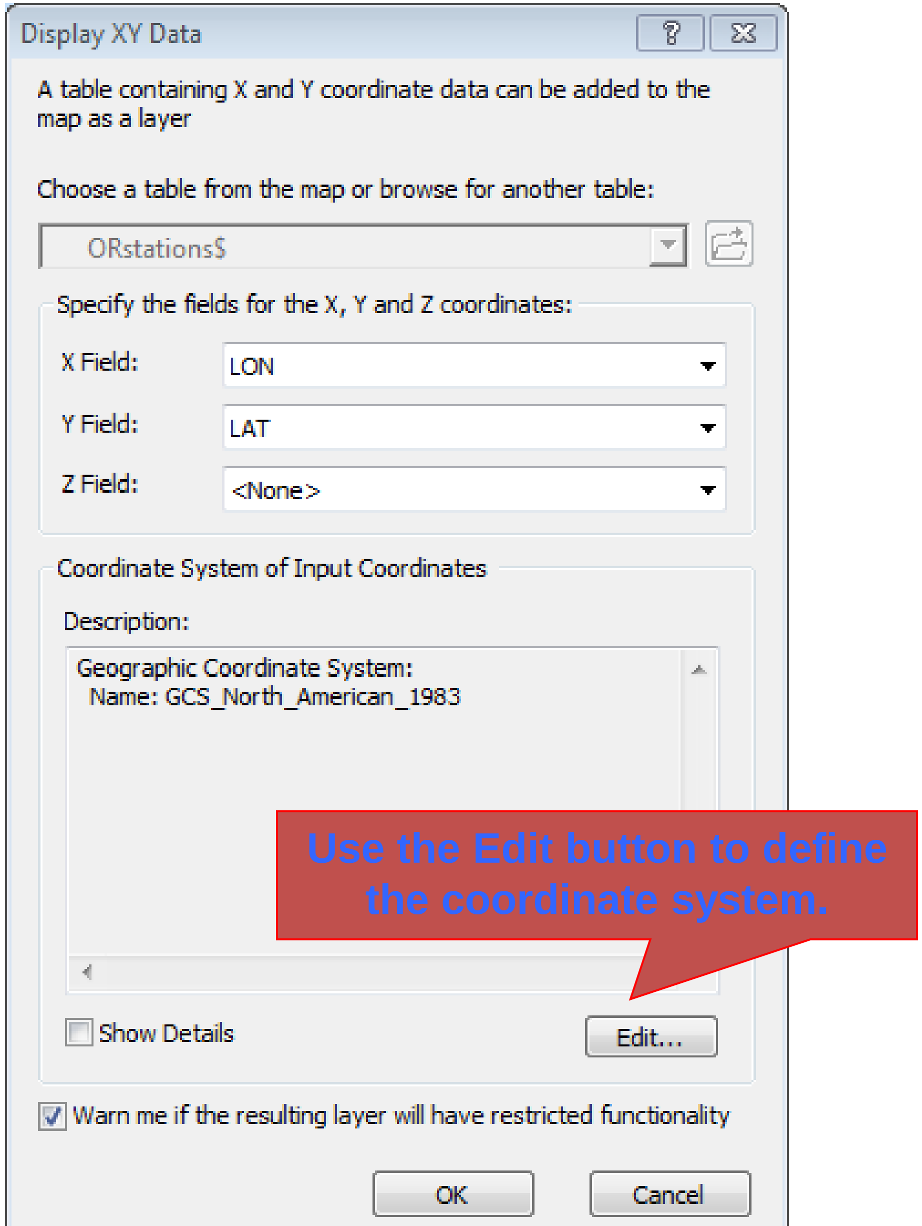
COOPID	LAT	LON	STATION I
350036	42.183333	-119.900000	ADEL
350118	42.966667	-120.000000	ALKALI LAKE
350145	44.400000	-123.750000	ALSEA F H FALL
350197	44.916667	-120.733333	ANTELOPE 1 NW
350217	42.250000	-123.166667	APPLEGATE

Geographic coordinate system (GCS)

A table with x-y values can be displayed as points.
Use any coordinate system, but you must know what it is.

Display XY Data

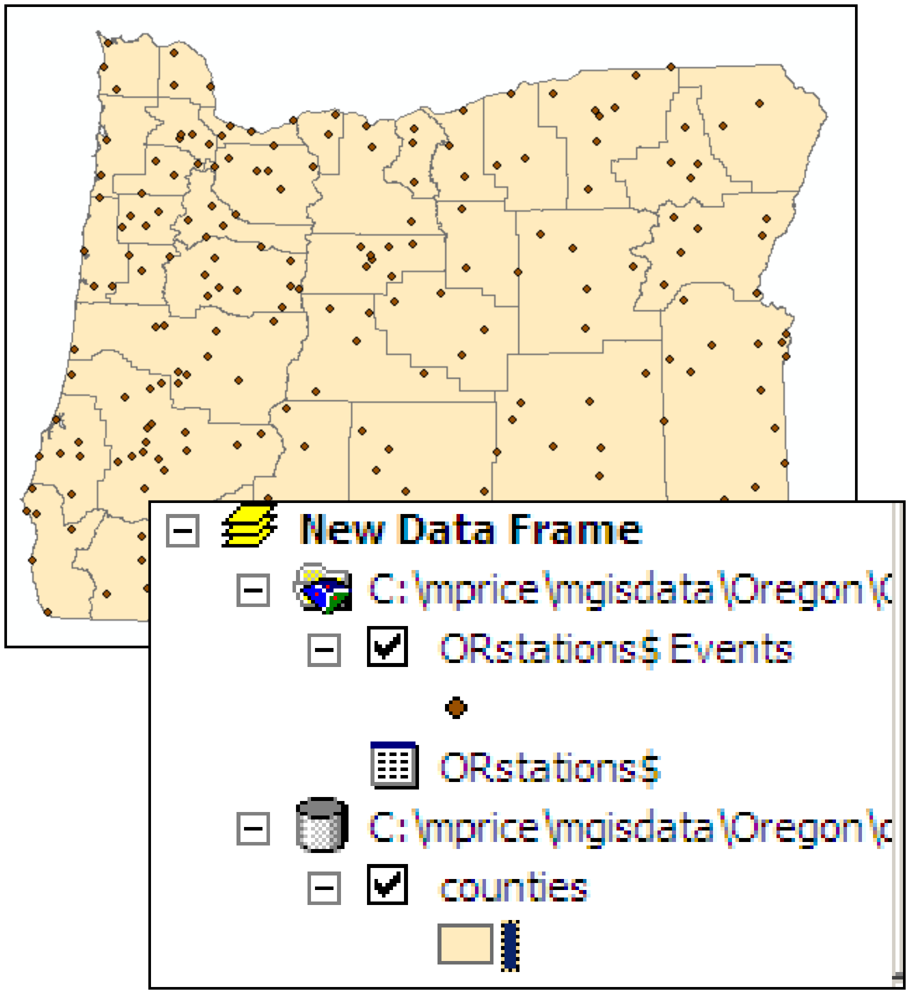




Degrees indicate a geographic coordinate system (GCS). NAD83 is the most common.

Other tables might have units in UTM meters, State Plane feet, etc. You must know the coordinate system to create correctly located points.

Oregon climate stations



- Stations appear as points on the map, an “event layer”
- Export to shapefile or feature class to keep permanently

Thank You